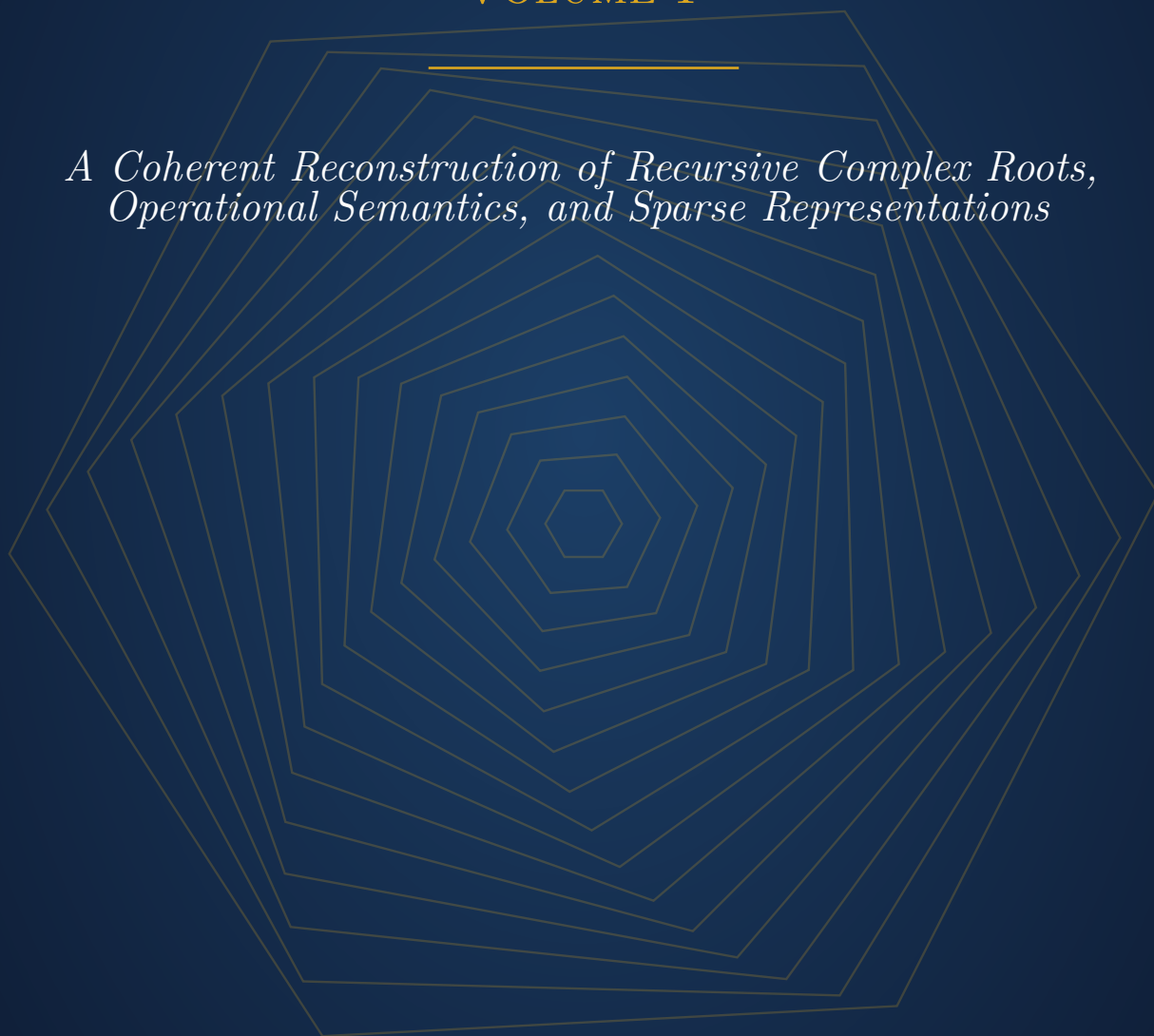


THE PULICKAL NOTEBOOK

VOLUME I

*A Coherent Reconstruction of Recursive Complex Roots,
Operational Semantics, and Sparse Representations*



PEARL BIPIN PULICKAL

The Pulickal Notebook

A Coherent Reconstruction of Recursive Complex Roots,
Operational Semantics, Sparse Representations, and Useful
Explorations

Pearl Bipin Pulickal

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Preface: What This Notebook Is Trying to Be

This notebook is not a collection of claims that have all survived scrutiny in their original form. It is a reconstruction. The guiding rule is simple:

A result is retained because it is correct, useful, structurally interesting, or diagnostically valuable. A result is not retained merely because it was once believed to be true.

The work collected here grew out of several independent experiments with recursive complex roots, complex-valued representations, probability amplitudes, logic tables, information measures, sparse numerical representations, and speculative physical interpretations. Some of those experiments turned into mathematics. Some turned into questions. Some turned into counterexamples to their own original claims. The purpose of this notebook is to put those objects into one place without pretending that they are all one theory.

There are two genuine mathematical cores.

First, there is a branch-sensitive theory of the recursion

$$z_{n+1}^r = -z_n, \quad r \geq 2,$$

with the principal branch and the full multivalued correspondence separated from each other. This theory is exact and convergent in the appropriate senses. The same algebraic relation supports several different limiting objects: a deterministic principal orbit, a finite-depth inverse polygon, a symbolic branch space, and a probability measure on branch histories. These are related but not identical.

Second, there is a task-semantic framework for information states and resource lower bounds. Given a deterministic specification and a specified family of future tasks, two states are equivalent exactly when all admissible future tasks give the same answer. The quotient by this relation is the canonical semantic state space. This gives a precise minimum-state theorem, after which resource statements must be made relative to an explicitly specified implementation architecture.

A third body of material is deliberately labeled as satellite mathematics. This includes a corrected three-outcome coin model, a complex-amplitude construction whose amplitude-squared set function has a quantum-measure interpretation when the underlying amplitude defines a complex measure, a real surrogate obtained from the Lambert W function, a complex-vector representation of ordinary arithmetic, a finite-valued logic algebra, a carefully stated Sleeping Beauty analysis, and a sparse signed pure-prime-power representation problem. These are preserved because they contain useful mathematics, not because the original framing was always defensible.

Several attractive phrases from earlier drafts are intentionally absent. In particular, this notebook does not claim that recursive imaginaries replace complex numbers, that randomness

is universally illusory, that topology admits lossless compression, that a single scalar measures mathematical rigor, or that a new notation automatically constitutes a new theory. Such claims are not needed for the mathematical results that survive.

Integrity note. The word *convergent* in this notebook is always qualified. A sequence may converge pointwise, in Hausdorff distance, weakly as a measure, or only after Cesàro averaging. A conjectural asymptotic statement is not called a convergence theorem. An open research programme is not presented as an established theory.

Reading Guide and Status Legend

The chapters are organized in increasing conceptual distance from the core recursion.

1. Chapters 1–9 develop the recursive-root core: branch conventions, exact principal dynamics, phase perturbations, inverse trees, Hausdorff geometry, branch measures, the polynomial bridge, infinite-depth limits, and logarithmic branch defects. Several sections deliberately rederive the same facts from different viewpoints so that the reader can see which conclusions depend on branch choice and which do not.
2. Chapters 10–13 develop the Pulickal Framework: operational semantics, exact realization, architecture-aware capacity, and approximate realization. The framework is formulated as a sequence of increasingly structured mathematical models rather than as a single universal information metric.
3. Chapter 14 develops the finite-height pure-prime-power representation problem and its conjectural asymptotics. Exact finite computation, counting lower bounds, height dependence, and the distinction between prime targets and all integers are treated separately.
4. Chapters 15–26 contain selected satellite investigations that survive after correction: complex amplitudes, the three-outcome coin, a Lambert W surrogate, realification, finite-valued logic, Sleeping Beauty, complex-coefficient oscillators, transform reductions, recursive probability, formal hyper-imaginary spaces, and task-dependent compression. Each satellite chapter states explicitly what is proved, what is merely modeled, and what has been discarded.
5. Chapters 27–30 are the synthesis layer: mathematical inventory, convergence map, research programme, and rules for extending the notebook without inflating its claims.
6. After the closing perspective, the document enters an appendix portion containing detailed rederivations, independent proof checks, exact-computation methodology, counterexamples, and literature-oriented context. The appendices are deliberately redundant: their purpose is auditability rather than narrative efficiency.

The status words are used literally:

Theorem-level	Proved in the notebook under clearly stated assumptions.
Proposition-level	A proved local structural fact, often useful inside a larger argument.
Conjectural	A precise statement whose truth is not established here.
Heuristic	A model or motivation that is useful for intuition but not a proof.
Exploratory	A mathematically well-defined construction retained primarily because it suggests questions.
Historical correction	A former claim is explicitly replaced by a stronger or more accurate statement.
Discarded	The original proposal is not retained as a mathematical claim.

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Notation at a Glance

Symbol	Meaning
i	the ordinary imaginary unit, $i^2 = -1$
$\text{Arg } z$	principal argument in $(-\pi, \pi]$
$\text{Log } z$	principal logarithm $\log z + i \text{Arg } z$
$\text{Root}_r^{\text{pr}}(z)$	principal r th root $\exp((1/r) \text{Log } z)$
F_r	principal map $F_r(z) = \text{Root}_r^{\text{pr}}(-z)$
$\mathcal{F}_r(z)$	full root correspondence $\{w : w^r = -z\}$
$\zeta_{\pm}$	$e^{\pm i\pi/(r+1)}$, the principal two-cycle endpoints
$A_n(z_0)$	depth- n inverse level under $w \mapsto -w^r$
ν_p	branch measure generated by branch probabilities $p = (p_0, \dots, p_{r-1})$
X, T, Y	information states, future tasks, and outputs in the Pulickal Framework
$\sigma_S(x)$	behavioral signature of x under specification S
$D(S)$	cardinality of the operational quotient
$\text{Cap}_{\mathcal{A}}(B)$	configuration capacity of an implementation class $\mathcal{A}$ under budget B
$\lambda_{k,H}(n)$	minimum signed pure-prime-power atomic length with height bound H
$M_k(X, H)$	prime-target extremal maximum with target bound X and height bound H
$M_k(B)$	diagonal specialization $M_k(B, B)$
$M_k^{\infty}(X)$	unrestricted prime-target extremal length
$N_k(X, H)$	all-integer extremal length under the same finite-height dictionary
$Q(A)$	amplitude-squared set function $ \nu(A) ^2$ in the corrected complex-amplitude discussion

Chapter 1

The Organizing Principle: One Equation, Several Objects

1.1 The algebraic seed

Fix an integer $r \geq 2$. The algebraic relation

$$w^r = -z \tag{1.1}$$

has r distinct solutions whenever $z \neq 0$. Therefore the phrase “iterate the r th root of $-z$ ” is incomplete until one says which root is being selected.

There are four distinct constructions that will matter throughout the notebook.

- (a) A *principal deterministic function* F_r chooses one root at each step.
- (b) A *multivalued correspondence* retains all r possible roots.
- (c) A *symbolic branch process* records the branch selected at each step.
- (d) A *probability measure on histories* puts a law on the symbolic branch space.

The mistake to avoid is to regard all four as different notations for the same limit. They answer different questions.

1.2 A useful hierarchy

The recursion provides a particularly clean hierarchy:

algebraic relation $\longrightarrow$ branch selection $\longrightarrow$ deterministic dynamics,

while retaining all branches instead gives

algebraic relation $\longrightarrow$ inverse correspondence $\longrightarrow$ tree $\longrightarrow$ symbolic boundary $\longrightarrow$ measure.

The deterministic and multivalued constructions share the same radial scaling law but differ completely in their angular organization.

Status: The organizing picture is mathematically sound. The notebook will therefore never use a single undefined symbol such as “ i_∞ ” without specifying which limiting object is meant.

1.3 How the notebook closes questions without forcing a single limit

A coherent notebook need not force every chapter to end at a single scalar constant. A better endpoint is an exact boundary between questions. In the recursive-root part, the endpoint is the distinction between principal orbit, inverse geometry, and branch measure. In the operational part, the endpoint is the distinction between semantic necessity and architecture-specific cost. In the exploratory part, the endpoint is a precise statement of what remains unknown.

This is the first principle of the notebook:

Do not manufacture convergence where the mathematics naturally produces several different limits. Instead, identify the correct object and prove convergence for that object whenever possible.

Chapter 2

Principal Branches and the Exact Negative-Root Map

2.1 Principal logarithm and principal root

For $z \in \mathbb{C} \setminus \{0\}$ define

$$\text{Arg } z \in (-\pi, \pi], \quad \text{Log } z = \log |z| + i \text{Arg } z.$$

For $r \geq 2$ define

$$\text{Root}_r^{\text{Pr}}(z) = \exp\left(\frac{1}{r} \text{Log } z\right), \quad \text{Root}_r^{\text{Pr}}(0) = 0. \quad (2.1)$$

The full root correspondence is

$$\mathcal{R}_r(z) = \{w \in \mathbb{C} : w^r = z\}.$$

Clearly $\text{Root}_r^{\text{Pr}}(z) \in \mathcal{R}_r(z)$, but the two objects are not equal.

2.2 Definition of the principal negative-root map

Set

$$F_r(z) = \text{Root}_r^{\text{Pr}}(-z). \quad (2.2)$$

The principal deterministic recursion is

$$z_{n+1} = F_r(z_n). \quad (2.3)$$

The corresponding multivalued system is

$$z_{n+1} \in \mathcal{F}_r(z_n), \quad \mathcal{F}_r(z) = \{w : w^r = -z\}. \quad (2.4)$$

2.3 Radial dynamics

Proposition 2.1. *For every $z \in \mathbb{C}$,*

$$|F_r(z)| = |z|^{1/r}. \quad (2.5)$$

Hence every principal orbit satisfies

$$|z_n| = |z_0|^{1/r^n}. \quad (2.6)$$

Proof. For $z \neq 0$,

$$F_r(z) = \exp\left(\frac{1}{r} \operatorname{Log}(-z)\right),$$

and therefore

$$|F_r(z)| = \exp\left(\frac{1}{r} \Re \operatorname{Log}(-z)\right) = \exp\left(\frac{1}{r} \log |z|\right) = |z|^{1/r}.$$

Iteration gives the formula. □

Writing $u_n = \log |z_n|$ gives the linear recurrence

$$u_{n+1} = \frac{1}{r} u_n, \quad u_n = \frac{u_0}{r^n}. \quad (2.7)$$

Hence

$$|z_n| \rightarrow 1 \quad (2.8)$$

for every $z_0 \neq 0$.

2.4 The actual angular map

Let $z = e^{i\theta}$ with $\theta \in (-\pi, \pi]$. The principal argument of $-z$ is

$$\operatorname{Arg}(-e^{i\theta}) = \begin{cases} \theta + \pi, & -\pi < \theta \leq 0, \\ \theta - \pi, & 0 < \theta \leq \pi. \end{cases} \quad (2.9)$$

Thus the principal angular map is

$$T_r(\theta) = \begin{cases} (\theta + \pi)/r, & -\pi < \theta \leq 0, \\ (\theta - \pi)/r, & 0 < \theta \leq \pi. \end{cases} \quad (2.10)$$

The principal-value reduction in (2.9) is essential. A lifted recurrence that omits it describes a different dynamical system.

2.5 Absolute-angle reduction

Define

$$\alpha_n = |\operatorname{Arg} z_n| \in [0, \pi].$$

Then (2.10) gives

$$\alpha_{n+1} = \frac{\pi - \alpha_n}{r}. \quad (2.11)$$

The unique fixed point is

$$\alpha_* = \frac{\pi}{r+1}. \quad (2.12)$$

Subtracting α_* from (2.11) yields

$$\alpha_{n+1} - \alpha_* = -\frac{1}{r}(\alpha_n - \alpha_*), \quad (2.13)$$

so

$$\boxed{\alpha_n = \frac{\pi}{r+1} + \left(-\frac{1}{r}\right)^n \left(\alpha_0 - \frac{\pi}{r+1}\right)}. \quad (2.14)$$

In particular $\alpha_n \rightarrow \pi/(r+1)$ at the exact rate r^{-n} .

2.6 The attracting two-cycle

Define

$$\zeta_+ = e^{i\pi/(r+1)}, \quad \zeta_- = e^{-i\pi/(r+1)}. \quad (2.15)$$

Theorem 2.2. *For every $r \geq 2$, $\{\zeta_+, \zeta_-\}$ is the unique nonzero periodic orbit of the principal map F_r . Every nonzero principal orbit has this pair as its ω -limit set.*

Proof. The radial recurrence forces every nonzero periodic point to have modulus 1, because a nonzero $u = \log |z|$ cannot be periodic under $u \mapsto u/r$ unless $u = 0$. For the angular magnitude, periodicity would require a periodic point of the affine contraction

$$A_r(\alpha) = \frac{\pi - \alpha}{r}.$$

Every iterate A_r^p has slope $(-1/r)^p$ and therefore exactly one fixed point. The unique fixed point of A_r is $\pi/(r+1)$, so there is exactly one periodic angular magnitude. The sign alternates under the principal map, giving precisely ζ_+ and ζ_- . Direct substitution verifies

$$F_r(\zeta_+) = \zeta_-, \quad F_r(\zeta_-) = \zeta_+.$$

Finally, the exact convergence of α_n and $|z_n|$ proves convergence of the even and odd subsequences to the two cycle points. $\square$

2.7 Exact distance to the moving cycle

For $n \geq 1$, let $\varepsilon_n \in \{-1, +1\}$ satisfy

$$\text{Arg } z_n = \varepsilon_n \alpha_n,$$

and define

$$\zeta_n = e^{i\varepsilon_n \alpha_*}.$$

Then the exact cosine-law identity is

$$|z_n - \zeta_n|^2 = \left(|z_0|^{1/r^n} - 1\right)^2 + 2|z_0|^{1/r^n} (1 - \cos(\alpha_n - \alpha_*)). \quad (2.16)$$

Since

$$|z_0|^{1/r^n} = 1 + \frac{\log |z_0|}{r^n} + O(r^{-2n})$$

and

$$\alpha_n - \alpha_* = \left(-\frac{1}{r}\right)^n (\alpha_0 - \alpha_*),$$

one obtains

$$|z_n - \zeta_n| = \frac{1}{r^n} \sqrt{(\log |z_0|)^2 + (\alpha_0 - \alpha_*)^2} + O(r^{-2n}). \quad (2.17)$$

Thus the principal orbit has an exact asymptotic rate after the moving two-cycle is chosen.

2.8 The square-root case

For $r = 2$,

$$\alpha_* = \frac{\pi}{3}, \quad \zeta_{\pm} = e^{\pm i\pi/3}. \quad (2.18)$$

Starting from $z_0 = 1$ gives the principal orbit

$$1, \quad e^{i\pi/2}, \quad e^{-i\pi/4}, \quad e^{3i\pi/8}, \quad e^{-5i\pi/16}, \dots$$

whose even and odd subsequences converge respectively to the two sixth-root-of-unity directions.

Warning 2.3. A lifted recurrence that writes $\theta_{n+1} = (\theta_n + \pi)/2$ for all n is not the principal dynamics. It omits principal-value reduction and therefore converges to a different formal angle. The corrected principal dynamics oscillates between two limiting angles.

2.9 Continuity domains and the derivative on a branch

The principal map is continuous on each region where the principal argument of $-z$ avoids the branch cut. The branch cut for Log is $(-\infty, 0]$; consequently the discontinuity set of $z \mapsto \text{Root}_r^{\text{pr}}(-z)$ is the preimage of that cut under $z \mapsto -z$, namely $(0, \infty)$. Although the branch formula changes across this ray, the assignment $F_r(0) = 0$ makes the map continuous at the origin. Thus one must distinguish local complex analyticity on a branch domain from global continuity of the principal map.

On any branch domain, differentiation of the algebraic identity

$$F_r(z)^r = -z$$

gives

$$F_r'(z) = \frac{F_r(z)}{rz}, \quad (2.19)$$

whenever $z \neq 0$ and the branch is locally analytic. Hence

$$|F_r'(z)| = \frac{1}{r|z|^{1-1/r}}. \quad (2.20)$$

In a neighborhood of the attracting two-cycle, where $|z|$ is close to 1, the local derivative magnitude is close to $1/r$. This local contraction is consistent with the exact r^{-n} asymptotic rate already obtained, but it should not be mistaken for a global contraction theorem on all of $\mathbb{C}$ because the principal map has a branch discontinuity and its derivative need not be small near 0.

Remark 2.4. The exact radial recurrence is stronger than a local derivative estimate: it gives the modulus at every iterate without any appeal to linearization.

2.10 Explicit sign dynamics

For a nonzero orbit write $z_n = \rho_n e^{i\theta_n}$ with $\theta_n \in (-\pi, \pi]$. If $0 < \theta_0 < \pi$, then the sign of the principal angle alternates

$$+, -, +, -, \dots,$$

whereas if $\theta_0 < 0$, it alternates

$$-, +, -, +, \dots$$

The special case $\theta_0 = 0$ begins with $\theta_1 = \pi/r > 0$ and then follows the same alternating pattern. The endpoint $\theta_0 = \pi$ is exceptional: it gives $\theta_1 = 0$, after which the orbit follows the usual pattern. Thus, away from these endpoint cases, the entire complex orbit can be reconstructed from three scalar pieces of data:

$$\rho_n = |z_0|^{1/r^n}, \quad \alpha_n = \frac{\pi}{r+1} + \left(-\frac{1}{r}\right)^n \left(\alpha_0 - \frac{\pi}{r+1}\right), \quad \varepsilon_{n+1} = -\varepsilon_n. \quad (2.21)$$

This makes the decomposition into radial contraction, angular contraction, and discrete sign switching completely explicit.

2.11 A quantitative two-cycle estimate

Set

$$a = \log |z_0|, \quad b = \alpha_0 - \alpha_*.$$

Then

$$\rho_n = e^{a/r^n}, \quad \alpha_n - \alpha_* = (-1)^n b/r^n.$$

Using the elementary estimates $|e^x - 1 - x| \leq Cx^2$ on bounded intervals and $1 - \cos y \leq y^2/2$, one obtains, for every fixed initial point and all sufficiently large n ,

$$\left| |z_n - \zeta_n| - \frac{\sqrt{a^2 + b^2}}{r^n} \right| \leq \frac{C(z_0, r)}{r^{2n}}. \quad (2.22)$$

The constant can be made explicit by fixing a neighborhood in which $|a|/r^n$ and $|b|/r^n$ are at most 1. The important point is structural: the radial and angular errors enter quadratically and therefore combine through a Euclidean norm rather than through a sum of absolute values.

2.12 What happens at the exceptional point

The radial convergence statement deliberately excludes $z_0 = 0$. At the origin, the principal recursion is stationary:

$$z_n = 0 \quad \text{for all } n.$$

Thus the origin is a fixed point, while every nonzero orbit approaches the unit circle in modulus and the attracting two-cycle in angle. The theorem about the unique nonzero periodic orbit does not conflict with the existence of the zero fixed point because the theorem explicitly excludes zero.

2.13 General lesson

The recursive operation is not intrinsically a single complex number at infinite depth. The broader subject of iterated complex radicals predates this notebook; convergence questions for recursively chosen complex square roots were studied by Ogilvy and Walker.[2, 1] For the principal negative-root family it naturally yields a two-cycle. A single scalar can be recovered by an additional operation, for example the Cesàro mean discussed next.

Proposition 2.5 (Cesàro limit of the principal orbit). *For $z_0 \neq 0$, the arithmetic mean of the principal iterates satisfies*

$$\frac{1}{N} \sum_{n=1}^N z_n \longrightarrow \frac{1}{2} (\zeta_+ + \zeta_-) = \cos \frac{\pi}{r+1}. \quad (2.23)$$

Proof. The even and odd subsequences converge to ζ_+ and ζ_- in some order, while the error is summable because it is $O(r^{-n})$. The Cesàro mean is therefore the average of the two limiting cycle points. $\square$

For $r = 2$ this gives

$$\frac{1}{N} \sum_{n=1}^N z_n \rightarrow \cos \frac{\pi}{3} = \frac{1}{2}.$$

This is a valid scalar limit, but it is not the ordinary pointwise limit of the sequence.

Chapter 3

Phase Perturbations and Piecewise-Affine Dynamics

3.1 The phase family

A natural deformation is

$$F_{r,\beta}(z) = \text{Root}_r^{\text{pr}}(e^{i\beta}z), \quad \beta \in \mathbb{R}. \quad (3.1)$$

Its radial behavior is unchanged:

$$|F_{r,\beta}(z)| = |z|^{1/r}. \quad (3.2)$$

The parameter β therefore modifies the angular system while leaving the logarithmic radial contraction intact.

3.2 Exact principal-angle formula

For $\theta \in (-\pi, \pi]$, let $\kappa_\beta(\theta) \in \mathbb{Z}$ be the unique integer for which

$$-\pi < \theta + \beta - 2\pi\kappa_\beta(\theta) \leq \pi. \quad (3.3)$$

Then

$$T_{r,\beta}(\theta) = \frac{\theta + \beta - 2\pi\kappa_\beta(\theta)}{r}. \quad (3.4)$$

The function κ_β is piecewise constant and changes only when $\theta + \beta$ crosses a principal-argument boundary. Thus each continuity branch is an affine contraction with slope $1/r$; the global map is generally discontinuous at the branch boundaries.

3.3 What can be claimed safely

The phase family is already a rich one-dimensional dynamical system. It is reasonable to ask about fixed points, periodic itineraries, parameter windows, and symbolic branch sequences. It is not necessary to claim that the family is a new dynamical-systems class. Contracted rotations and piecewise-affine contractions are established subjects.

The safe notebook-level statement is therefore:

The recursive complex-root problem provides an explicit, branch-sensitive realization of a piecewise-affine contraction on a principal angular interval. This realization is useful because the branch index is visible and computable.

3.4 Periodic itineraries as finite integer data

The phase-perturbed map can be analyzed itinerary by itinerary. Suppose a period- p orbit follows branch indices $\kappa_0, \dots, \kappa_{p-1}$. Then its lifted recurrence is

$$\theta_{j+1} = \frac{\theta_j + \beta - 2\pi\kappa_j}{r}, \quad j = 0, \dots, p-1. \quad (3.5)$$

Iterating gives

$$\theta_p = r^{-p}\theta_0 + \beta \sum_{m=1}^p r^{-m} - 2\pi \sum_{j=0}^{p-1} \kappa_j r^{-(p-j)}. \quad (3.6)$$

Imposing $\theta_p = \theta_0$ produces the unique candidate

$$\theta_0 = \frac{\beta \sum_{m=1}^p r^{-m} - 2\pi \sum_{j=0}^{p-1} \kappa_j r^{-(p-j)}}{1 - r^{-p}}. \quad (3.7)$$

This formula does not by itself prove that the itinerary occurs: the candidate must still satisfy all branch inequalities used to define the integers κ_j . The value of the formula is therefore diagnostic. It reduces the search for periodic orbits of a prescribed symbolic type to a finite collection of linear inequalities in the candidate angle and parameter β .

For a fixed admissible itinerary, the derivative of the p th return map is r^{-p} , so any such periodic orbit is locally attracting within its continuity itinerary. What remains nontrivial is the global combinatorics: which itineraries are admissible, when two itineraries collide at a branch boundary, and how the answer changes with β .

3.5 A useful distinction between local and global contraction

On each continuity interval, $T_{r,\beta}$ is affine with slope $1/r$. Consequently, if two points remain in the same branch itinerary for n steps, their angular separation is exactly divided by r^n . This is a strong local statement. It does not imply that arbitrary pairs of initial conditions become exponentially close under the global map, because a pair may encounter different branch cuts. In fact, the discontinuity can create an $O(1)$ separation in one step even when the initial angles are arbitrarily close on opposite sides of a cut.

This distinction is a recurring principle in the notebook: a formula can exhibit contraction on every smooth piece while the globally defined piecewise map remains discontinuous.

3.6 A branch-index viewpoint

A useful decomposition is

$$\theta_{n+1} = \frac{\theta_n + \beta}{r} - \frac{2\pi}{r} \kappa_\beta(\theta_n). \quad (3.8)$$

Thus the continuous contraction and the discrete branch event are separated. This separation is one of the most reusable pieces of the notebook: a continuous state variable may contract while an integer-valued branch record carries the nontrivial combinatorics.

Question 3.1. For fixed r and β , classify all periodic itineraries of the piecewise-affine map (3.4). Which rational rotation structures occur? Which itineraries are structurally stable under changes in β ? Which endpoint conventions change only symbolic coding rather than the underlying attractor?

These questions are recorded as open work rather than silently imported from a different literature.

Chapter 4

The Full Multivalued Correspondence and Inverse Tree

4.1 The forward polynomial

Define

$$P_r(z) = -z^r. \quad (4.1)$$

Then the multivalued root step is exactly the inverse relation of P_r .

4.2 Exact inverse levels

Let $A_n(z_0)$ be the set of points w satisfying

$$P_r^n(w) = z_0. \quad (4.2)$$

Since

$$P_r^n(w) = (-1)^{1+r+\dots+r^{n-1}} w^{r^n},$$

write

$$S_n = 1 + r + \dots + r^{n-1} = \frac{r^n - 1}{r - 1}.$$

Then the equation becomes

$$w^{r^n} = (-1)^{S_n} z_0. \quad (4.3)$$

If

$$\phi_n = \text{Arg}((-1)^{S_n} z_0), \quad \rho_n = |z_0|^{1/r^n},$$

then

$$A_n(z_0) = \left\{ \rho_n \exp \left(\frac{i(\phi_n + 2\pi j)}{r^n} \right) : 0 \leq j < r^n \right\}. \quad (4.4)$$

Proposition 4.1. *For $z_0 \neq 0$, the level $A_n(z_0)$ contains exactly r^n points, all on the circle of radius ρ_n , and the angular spacing is exactly $2\pi/r^n$.*

The geometry is therefore an exact regular polygon, rotated by a phase determined by $(-1)^{S_n} z_0$.

4.3 The rooted history tree

Every inverse step from a nonzero point creates r distinct predecessors. Recording the branch index $k \in \{0, 1, \dots, r-1\}$ at each level produces a rooted r -ary tree. A finite branch history is a word

$$(k_1, \dots, k_n) \in \{0, \dots, r-1\}^n.$$

The tree is a combinatorial object; the map from words to terminal complex numbers is a coding map.

The important distinction is:

Object	What it remembers
Principal orbit	One selected branch at every step
Inverse level	All terminal values at one depth
History tree	All finite branch sequences
Infinite symbolic space	Entire infinite branch histories
Pushforward measure	Statistical weight of histories

4.4 Endpoint and branch-label issues

The finite inverse level $A_n(z_0)$ is a set of r^n geometric points, whereas a branch-history tree contains r^n words of length n . For $z_0 \neq 0$ these cardinalities agree, but the identification depends on a branch-label convention. When branch labels are defined as exact algebraic inverse choices, the r^n branch histories are in bijection with the r^n points of $A_n(z_0)$. If one instead encodes branches by closed angular intervals, endpoint conventions can create duplicate symbolic codings of boundary points. The geometric set $A_n(z_0)$ is canonical, while such interval-based symbolic labels are convention-dependent.

This is the correct place to use the phrase “monodromy” carefully. The nontrivial bookkeeping arises because choosing roots globally around a closed loop forces one to remember how the argument winds. The notebook does not need a full monodromy theory to use the phenomenon: the exact formula for $A_n(z_0)$ already captures the finite-depth consequence.

4.5 Nested inverse formulas

Although the terminal set has the compact closed form above, an individual branch word can also be expanded recursively. If the chosen k_j is interpreted relative to a fixed angular representative at step j , then

$$\theta_{j+1} = \frac{\theta_j + \pi + 2\pi k_{j+1}}{r} \quad (4.5)$$

with the appropriate wrap convention. Repeated substitution yields an affine combination of the initial angle and the branch digits with coefficients that decay geometrically. This makes the inverse tree a contracting symbolic coding: changing a digit early in the history has a smaller effect on the terminal angle than changing a late digit, once a compatible lift convention has been fixed.

For geometric questions, however, the closed regular-polygon formula is preferable because it avoids the bookkeeping ambiguities of individual branch representatives. The notebook will therefore use the tree language for combinatorial questions and the polygon formula for metric questions.

4.6 Connection with the classical power map

The map $P_r(z) = -z^r$ is linearly conjugate to z^r : choose $c \neq 0$ with $c^{r-1} = -1$ and write

$$P_r(cz) = cz^r.$$

Thus the standard theory of the power map applies after a trivial rotation. The inverse tree and its monodromy interpretation are consequently classical in broad outline. What is worth preserving in this notebook is the explicit branch-word parametrization and the exact finite-level formulas.

Chapter 5

Hausdorff Convergence of Inverse Polygons

5.1 The finite polygons

Let $N_n = r^n$ and let $\rho_n = |z_0|^{1/N_n}$. The inverse level is a regular N_n -gon on radius ρ_n . Let S^1 denote the unit circle.

5.2 Exact Hausdorff distance

For a compact set pair $E, F \subset \mathbb{C}$ write

$$d_H(E, F) = \max \left\{ \sup_{x \in E} \text{dist}(x, F), \sup_{y \in F} \text{dist}(y, E) \right\}.$$

The maximum distance from a unit-circle point to the nearest vertex of the regular N_n -gon occurs at an angular midpoint. Thus

$$\sup_{|z|=1} \text{dist}(z, A_n(z_0)) = \sqrt{1 + \rho_n^2 - 2\rho_n \cos \frac{\pi}{N_n}}. \quad (5.1)$$

The distance from a polygon vertex of radius ρ_n to the unit circle is $|1 - \rho_n|$. The first quantity dominates, so

Theorem 5.1. *For every $z_0 \neq 0$ and every $n \geq 1$,*

$$\boxed{d_H(A_n(z_0), S^1) = \sqrt{1 + \rho_n^2 - 2\rho_n \cos(\pi/r^n)}}. \quad (5.2)$$

Moreover,

$$d_H(A_n(z_0), S^1) = \frac{\sqrt{(\log |z_0|)^2 + \pi^2}}{r^n} + O(r^{-2n}). \quad (5.3)$$

Proof. Let $N = r^n$. The angular distance to the nearest vertex is at most π/N and is exactly π/N at an arc midpoint. The nearest-point distance formula gives the first expression. For the asymptotic, expand

$$\rho = \exp((\log |z_0|)/N) = 1 + \frac{\log |z_0|}{N} + O(N^{-2})$$

and

$$\cos(\pi/N) = 1 - \frac{\pi^2}{2N^2} + O(N^{-4}).$$

Substitution gives

$$d_H^2 = \frac{(\log |z_0|)^2 + \pi^2}{N^2} + O(N^{-3}),$$

and taking square roots yields the result. □

5.3 Interpretation

This theorem is one of the cleanest examples in the notebook of different notions of convergence aligning. The radial error and angular mesh error are independent to first order, and their Euclidean combination produces a single asymptotic constant.

For z_0 on the unit circle, the radial term vanishes and the leading constant is simply π .

Chapter 6

Symbolic Measures, Uniform Branching, and Dimension

6.1 The symbolic space

Let

$$\Sigma_r = \{0, 1, \dots, r-1\}^{\mathbb{N}}$$

with its product topology. A Bernoulli law is specified by probabilities

$$p_0, \dots, p_{r-1} \geq 0, \quad \sum_{k=0}^{r-1} p_k = 1.$$

The natural inverse branches are most cleanly described in an angular coordinate rather than by calling them Euclidean similarities of the circle. Choose a cut interval, for example $I = (-\pi, \pi]$, and represent a point of S^1 by $e^{i\theta}$ with $\theta \in I$. Each inverse branch has an affine lift of the form

$$g_k(\theta) = \frac{\theta + \pi + 2\pi k}{r}, \quad k = 0, \dots, r-1, \quad (6.1)$$

followed, when necessary, by the chosen endpoint convention for the angular representative. On every branch interval the derivative is exactly $1/r$. Thus the contraction statement is an angular-coordinate statement (or, equivalently, a statement about the symbolic coding metric), not a claim that a globally defined Euclidean similarity of the entire circle exists. This distinction is important because a global continuous single-valued r th-root branch on S^1 does not exist. The branch maps instead select r arcs whose interiors are disjoint, with only endpoint identifications.

For later dimension statements, it is useful to keep three spaces distinct: the symbolic space Σ_r , the angular coding interval(s), and the geometric circle. The symbolic Bernoulli measure is unambiguous on Σ_r ; the geometric measure is its pushforward under the branch coding map.

6.2 Uniform branch selection

Under uniform weights $p_k = 1/r$, the depth- n inverse level consists of equally spaced points on the circle of radius

$$\rho_n = |z_0|^{1/r^n}.$$

The associated empirical measure is

$$\mu_n = \frac{1}{r^n} \sum_{z \in A_n(z_0)} \delta_z.$$

Theorem 6.1 (Weak convergence of uniform inverse levels). *For every $z_0 \neq 0$ and every bounded continuous function $f : \mathbb{C} \rightarrow \mathbb{C}$,*

$$\int f d\mu_n \longrightarrow \int_{S^1} f dm_{S^1}. \quad (6.2)$$

Proof. The radius satisfies $\rho_n \rightarrow 1$. All sufficiently large inverse levels lie in a fixed compact annulus containing S^1 , and f is uniformly continuous there. Hence replacing the radius ρ_n by 1 changes the empirical average by a quantity tending to zero. After this replacement, the points are equally spaced with mesh $2\pi/r^n$, so the resulting averages are Riemann sums for normalized arc length on S^1 . $\square$

Remark 6.2. If $z_0 \in S^1$, then $\rho_n = 1$ for every n . The formulation with $f \in C_b(\mathbb{C})$ is used to avoid the domain mismatch that would occur if f were defined only on S^1 while μ_n were supported on a nearby circle.

6.3 A quantitative uniform-measure estimate

The weak-convergence theorem can be strengthened for Lipschitz test functions. Let f be Lipschitz on an annulus containing all sufficiently large inverse levels and S^1 , with Lipschitz constant L . Couple each vertex $\rho_n e^{i\vartheta_j}$ with the point $e^{i\vartheta_j}$ on the unit circle. Then

$$|\rho_n e^{i\vartheta_j} - e^{i\vartheta_j}| = |\rho_n - 1|. \quad (6.3)$$

The remaining difference between the uniform average over equally spaced angles and the Haar integral is a Riemann-sum error of order $O(L/r^n)$. Hence

$$\left| \int f d\mu_n - \int_{S^1} f dm_{S^1} \right| \leq L |\rho_n - 1| + O\left(\frac{L}{r^n}\right). \quad (6.4)$$

Since $|\rho_n - 1| = O(r^{-n})$, the overall discrepancy is $O(r^{-n})$ for Lipschitz observables.

6.4 Why the entropy quotient is the natural dimension

A more explicit proof outline helps clarify exactly what is being used. Give the symbolic space the ultrametric

$$d_{\text{sym}}(\omega, \eta) = r^{-(m(\omega, \eta)-1)},$$

where $m(\omega, \eta)$ is the first index at which the two sequences differ. With this convention, a length- n cylinder has diameter r^{-n} in the symbolic metric. Under the branch coding, the corresponding angular interval has length proportional to r^{-n} .

For a Bernoulli sequence with positive probabilities, the strong law of large numbers implies

$$-\frac{1}{n} \log \nu_p([\omega_1 \cdots \omega_n]) \rightarrow H(p)$$

for almost every ω . At the same time, the logarithm of the geometric diameter divided by n approaches $-\log r$. Therefore the local dimension is

$$\lim_{n \rightarrow \infty} \frac{\log \nu_p([\omega_1 \cdots \omega_n])}{\log r^{-n}} = \frac{H(p)}{\log r}$$

for almost every code. In the separated regular coding, this almost-everywhere local dimension is the Hausdorff dimension of the measure.

This argument also makes the sign convention transparent. The Lyapunov exponent is negative,

$$\chi = \log(1/r),$$

while the positive contraction rate appearing in the denominator is $-\chi = \log r$.

6.5 Biased branch measures

For nonuniform $p = (p_0, \dots, p_{r-1})$, the symbolic Bernoulli measure pushes forward through the explicit r -adic branch coding to a self-similar probability measure on S^1 . The cylinder intervals have contraction ratio $1/r$ and disjoint interiors (up to endpoint identifications). The Shannon entropy of the symbol distribution is

$$H(p) = - \sum_{k=0}^{r-1} p_k \log p_k. \quad (6.5)$$

The geometric contraction factor is $1/r$, so the logarithmic contraction exponent is

$$\chi = \log(1/r) = -\log r. \quad (6.6)$$

It is therefore $|\chi| = \log r$, rather than $\log r$ itself, that appears in the denominator of the usual entropy-over-contraction formula. In the regular r -adic coding, and with endpoint identifications handled in the standard way, the pushforward Bernoulli measure has measure dimension

$$\boxed{\dim_H \nu_p = \frac{H(p)}{\log r}.} \quad (6.7)$$

Here $\dim_H \nu_p$ means the Hausdorff dimension of the measure, not the Hausdorff dimension of its support. The support can be all of S^1 while the measure dimension is strictly smaller than 1 for nonuniform positive weights. For the uniform law, $H(p) = \log r$, so the measure dimension is 1. The formula is the standard entropy-dimension relation for the separated regular r -adic coding; the present notebook uses it in that precise coordinate/coding sense rather than asserting a globally defined Euclidean IFS on S^1 . [6, 7]

Warning 6.3. Equation (6.7) is not an original law of “recursive probability.” It is an instance of standard self-similar measure theory applied to an explicit inverse-root system.

6.6 Chronological randomness is a different object

A random branch can be selected sequentially along one orbit, producing a random affine recursion for the angle. That process should not be conflated with a random point sampled from the limiting branch measure. The first is a chronological stochastic dynamical system; the second is a boundary measure on branch histories.

This distinction is important whenever one asks about expectations, almost-sure orbit behavior, or empirical distributions.

Chapter 7

The Polynomial Bridge: Julia Sets and Potential Theory

7.1 Why the Julia statement must be attached to the polynomial

The principal root map F_r is not a globally holomorphic self-map of the Riemann sphere because of branch selection. A standard Julia-set theorem should therefore be attached to the polynomial

$$P_r(z) = -z^r,$$

not to the principal inverse branch itself.

The Julia set of $z \mapsto z^r$ is the unit circle. Since P_r is a rotation-conjugate of z^r , its Julia set is also

$$J(P_r) = S^1.$$

The filled Julia set is the closed unit disk.

7.2 Green function

For the power map the escape-rate function has the explicit form

$$G(z) = \max\{0, \log |z|\}. \tag{7.1}$$

It satisfies

$$G(P_r(z)) = rG(z).$$

The logarithmic radial coordinate that contracts under inverse root iteration is exactly the coordinate that expands under the polynomial forward map.

7.3 Equilibrium measure

The normalized angular measure on S^1 is the equilibrium measure of the unit disk. For $z_0 \neq 0$, it also arises as the weak limit of the uniform inverse-level distributions. The point $z_0 = 0$ is exceptional: all inverse levels are $\{0\}$, so their empirical measures are δ_0 . This gives a clean bridge between explicit polygonal geometry and classical potential theory.

Integrity note. The meaningful statement is not that “recursive imaginaries discovered the Julia set.” The meaningful statement is that the inverse levels of the

explicit polynomial $-z^r$ approach its classical Julia set in an exactly computable way.

Chapter 8

Infinite Depth: What the Expression Can Mean

The phrase “infinite nested radical” can refer to several inequivalent limiting procedures. The notebook separates them.

8.1 Pointwise principal limit

For the principal negative-root recursion and $z_0 \neq 0$, the sequence z_n generally does not converge. Instead,

$$z_{2n} \rightarrow \zeta_{\text{even}}, \quad z_{2n+1} \rightarrow \zeta_{\text{odd}},$$

with the two values equal to ζ_+ and ζ_- in some order.

Thus the pointwise limit fails, while the two subsequential limits exist.

8.2 Cesàro limit

The average of the iterates converges to

$$\cos \frac{\pi}{r+1}.$$

This is a perfectly good infinite-depth scalar if the averaging operation is part of the definition.

8.3 Inverse-set limit

The inverse levels converge in Hausdorff distance to S^1 . This is not a point limit at all; it is a compact-set limit.

8.4 Measure limit

Uniform branch selection gives weak convergence to Haar probability on S^1 . Biased branch selection gives a self-similar measure whose dimension is controlled by entropy.

8.5 The right definition table

Procedure	Limit object	Status
One principal orbit	two-cycle	theorem
Cesàro averaging	real scalar $\cos(\pi/(r+1))$	theorem
All inverse points	unit circle	theorem
Uniform inverse measure	Haar measure	theorem
Biased inverse measure	self-similar measure	theorem under separation
Unspecified “infinite radical”	not well-defined	reject

Chapter 9

Branch Defects and Complex Logarithms

9.1 The source of the difficulty

For complex logarithms, identities that are automatic over the positive reals may acquire integer multiples of $2\pi i$. The principal logarithm satisfies

$$\text{Log}(ab) = \text{Log } a + \text{Log } b - 2\pi i k(a, b)$$

for an integer-valued defect $k(a, b)$ determined by the chosen principal arguments.

A convenient definition is

$$k(a, b) = \frac{\text{Log } a + \text{Log } b - \text{Log}(ab)}{2\pi i} \in \mathbb{Z}. \quad (9.1)$$

The defect is nonzero precisely when the principal-angle sum leaves the principal strip.

9.2 Complex powers

For a principal complex power $z^s = e^{s \text{Log } z}$, product factorization must retain branch information. A safe rule is:

Never simplify a principal logarithmic identity by treating Log as an algebraic homomorphism.

9.3 A corrected tangent-binomial perspective

Expressions such as $(1 + ix)^n$ can be expanded algebraically in the ordinary complex plane. What fails is not the binomial theorem but the attempt to replace a principal logarithm of a product by an unrestricted sum of principal logarithms. The correct repair is to track the integer winding defect.

This is one of the most useful conceptual lessons from the recursive-root work: branch indices are not decorative labels. They are the discrete data required to make principal formulas correct.

Chapter 10

The Pulickal Framework: Information States and Tasks

10.1 The central question

A specification may become much more elaborate without requiring more persistent information. What matters is not textual length but the set of future behaviors that must remain distinguishable.

Let X be a nonempty set of information states, let T be a nonempty family of future tasks, and let Y be an outcome space.

Definition 10.1. A deterministic specification is a map

$$F_S : X \times T \rightarrow Y. \quad (10.1)$$

For $x \in X$, its behavioral signature is

$$\sigma_S(x) = F_S(x, \cdot) \in Y^T, \quad (10.2)$$

Define the behavioral signature space by

$$\Sigma_S := \sigma_S(X) \subseteq Y^T. \quad (10.3)$$

Definition 10.2. For $x, y \in X$, define operational equivalence by

$$x \sim_S y \iff F_S(x, t) = F_S(y, t) \text{ for every } t \in T. \quad (10.4)$$

Proposition 10.3. $\sim_S$ is an equivalence relation.

Proof. Reflexivity and symmetry are immediate from equality. For transitivity, if $x \sim_S y$ and $y \sim_S z$, then for every $t \in T$,

$$F_S(x, t) = F_S(y, t) = F_S(z, t).$$

Hence $x \sim_S z$. □

10.2 The canonical quotient

Define

$$D(S) = \text{card}(X / \sim_S) \quad (10.5)$$

when the quotient is finite.

This is not a generic measure of information. It does not count pages, equations, symbols, proof steps, or memory cells. It counts the number of task-relevant behavioral classes.

10.3 The coarsest-sufficient representation theorem

Definition 10.4. A representation $h : X \rightarrow H$ is semantically sufficient if

$$h(x) = h(y) \implies \sigma_S(x) = \sigma_S(y).$$

Theorem 10.5. *The quotient map*

$$q_S : X \rightarrow X / \sim_S$$

is semantically sufficient and is minimal among exact semantically sufficient representations. More precisely, for every semantically sufficient $h : X \rightarrow H$ there is a unique map

$$\gamma : \text{range}(h) \rightarrow \Sigma_S$$

with

$$\gamma(h(x)) = \sigma_S(x),$$

and therefore, whenever $D(S) < \infty$,

$$\text{card}(\text{range}(h)) \geq D(S). \quad (10.6)$$

Proof. If $q_S(x) = q_S(y)$ then $x \sim_S y$, so their signatures agree. Thus q_S is sufficient. For a general sufficient h , define $\gamma(h(x)) = \sigma_S(x)$. If $h(x) = h(y)$ then semantic sufficiency gives $\sigma_S(x) = \sigma_S(y)$, so γ is well-defined. Surjectivity onto Σ_S is immediate from the definition of the signature space, and hence $\text{card}(\text{range}(h)) \geq \text{card}(\Sigma_S) = D(S)$. $\square$

10.4 A finite example

Let $X = \{0, 1, 2, 3\}$ and let the task family contain two tasks, t_1 and t_2 , with outputs in $\{0, 1\}$. Suppose

$$F_S(x, t_1) = \begin{cases} 0, & x \in \{0, 1\}, \\ 1, & x \in \{2, 3\}, \end{cases} \quad F_S(x, t_2) = \begin{cases} 0, & x \in \{0, 2\}, \\ 1, & x \in \{1, 3\}. \end{cases}$$

Then the signatures are

$$\sigma_S(0) = (0, 0), \quad \sigma_S(1) = (0, 1), \quad \sigma_S(2) = (1, 0), \quad \sigma_S(3) = (1, 1),$$

so all four states are operationally distinguishable and $D(S) = 4$. Any exact state representation must therefore have at least four persistent configurations. If the task family is reduced to only t_1 , then the quotient has two classes, $\{0, 1\}$ and $\{2, 3\}$, so the semantic state requirement falls to two configurations. The specification has not become “less truthful”; fewer distinctions are simply being asked of the representation.

This example is elementary, but it shows why the task family is a mathematical parameter of the framework rather than a footnote.

10.5 Why this stops the semantic regress

One may be tempted to ask for a deeper quantity measuring how complicated $D(S)$ is, then a deeper quantity measuring that quantity, and so on. That tower is optional, not required. Once the admissible tasks are fixed, the quotient answers the precise semantic question:

Which states are distinguishable by any task the model permits?

That is the intended stopping point. New questions about energy, communication, latency, noise, or probabilistic behavior are not “unfinished rigor”; they are different problems.

Status: The framework is intentionally convergent at the semantic layer.

The quotient is a complete answer to exact deterministic task equivalence. Further layers are added only when a new resource or new semantic regime is explicitly introduced.

Chapter 11

Exact Realization and State Separation

11.1 Task enrichment

Suppose $T_1 \subseteq T_2$, and write $D(S; T)$ for the quotient cardinality obtained when the admissible task family is T . Equality on the larger task family implies equality on the smaller family, hence

$$D(S; T_2) \geq D(S; T_1). \quad (11.1)$$

whenever both are finite.

A refinement may therefore create more distinctions. But that does not by itself prove a larger engineering cost.

11.2 Implementation model

An exact implementation I consists of a configuration set Z_I , a configuration map

$$\phi_I : X \rightarrow Z_I,$$

and a response map

$$G_I : Z_I \times T \rightarrow Y.$$

Exact realization means

$$G_I(\phi_I(x), t) = F_S(x, t) \quad \text{for all } x, t. \quad (11.2)$$

Theorem 11.1 (State-separation theorem). *If I exactly realizes S , then*

$$x \not\sim_S y \implies \phi_I(x) \neq \phi_I(y). \quad (11.3)$$

Consequently, if $D(S) < \infty$,

$$\text{card}(Z_I) \geq D(S). \quad (11.4)$$

Proof. If $x \not\sim_S y$, choose $t \in T$ with $F_S(x, t) \neq F_S(y, t)$. If $\phi_I(x) = \phi_I(y)$, exact realization gives

$$F_S(x, t) = G_I(\phi_I(x), t) = G_I(\phi_I(y), t) = F_S(y, t),$$

a contradiction. □

11.3 Binary memory as a corollary

If the implementation has b bits of persistent binary state, then $\text{card}(Z_I) \leq 2^b$, so

$$2^b \geq D(S) \tag{11.5}$$

and therefore

$$\boxed{b \geq \lceil \log_2 D(S) \rceil.} \tag{11.6}$$

The logarithm here is not the conceptual heart of the theorem. It is simply the capacity formula of a b -bit memory.

11.4 What the theorem does not prove

The state-separation theorem does not prove that every additional semantic distinction costs one physical bit, one transistor, one joule, or one unit of latency. It proves a lower bound on the number of persistent configurations in the specified state-based model.

To turn this into an engineering statement, one must specify the architecture and its resource law.

Chapter 12

Architecture-Aware Capacity and Resource Bounds

12.1 Capacity

Let $\mathcal{A}$ be a class of implementations, let $C(I)$ be a chosen scalar cost functional on implementations, and let B be a resource budget. Define abstractly

$$\text{Cap}_{\mathcal{A}}(B) := \sup_{I \in \mathcal{A}: C(I) \leq B} \text{card}(Z_I)$$

to be the supremal number of persistent configurations that the class can provide under budget B . If the supremum is attained, it is a maximum. Using a supremum avoids an unnecessary attainment assumption.

Then

Theorem 12.1 (Capacity impossibility). *If*

$$D(S) > \text{Cap}_{\mathcal{A}}(B), \tag{12.1}$$

then no exact realization of S exists in architecture class $\mathcal{A}$ under budget B .

Proof. An exact implementation needs at least $D(S)$ configurations by state separation, while every implementation in the class under budget B supplies at most $\text{Cap}_{\mathcal{A}}(B)$. If the former exceeds the latter, exact realization is impossible. $\square$

12.2 Conditionality is a strength, not a weakness

The theorem is intentionally conditional. It says nothing about architectures outside $\mathcal{A}$, or about an omitted communication channel, a remote database, recomputation, or a new sensing operation. Those must be included in the model if they are allowed.

12.3 Multiple resources

Suppose an implementation has a cost vector

$$C(I) = (C_1(I), \dots, C_m(I)).$$

The natural object is then the Pareto frontier of achievable costs, not a universal scalar. A theorem may establish a lower bound in one coordinate while leaving the trade-off between coordinates open.

12.4 No universal monotonic cost law

A refinement can increase $D(S)$ without increasing the optimal cost if the new distinction can be generated by recomputation, external information, compression, or a better architecture. Let

$$C^*(S) := \inf\{C(I) : I \text{ exactly realizes } S\}$$

for whatever scalar cost functional C has been chosen. Therefore the implication

$$D(S') > D(S) \implies C^*(S') > C^*(S)$$

is false in general.

The framework is strongest when it distinguishes what follows from semantics alone from what requires architecture-specific assumptions.

Chapter 13

Approximate Realization: Packing and Covering

13.1 Why exact cardinality is insufficient for approximation

For approximate realization, suppose the behavioral signature space Σ_S carries a metric d . The implementation may store a representative whose signature is within worst-case error ε of the true signature.

Definition 13.1. The δ -packing number is

$$\text{Pack}(\Sigma, \delta) = \sup\{|E| : E \subseteq \Sigma, d(x, y) > \delta \text{ for all distinct } x, y \in E\}.$$

Definition 13.2. The ε -covering number is

$$\text{Cov}(\Sigma, \varepsilon) = \inf \left\{ |E| : E \subseteq \Sigma, \Sigma \subseteq \bigcup_{x \in E} \overline{B}(x, \varepsilon) \right\}.$$

Definition 13.3. Let $K^*(\varepsilon)$ denote the infimum of the numbers of persistent configurations among implementations in the specified approximation model having worst-case behavioral error at most ε . When an optimum is attained, this infimum is a minimum.

13.2 Packing obstruction

Theorem 13.4. Suppose an implementation has at most K configurations and every signature is reproduced with worst-case error at most ε . Then

$$K \geq \text{Pack}(\Sigma_S, 2\varepsilon). \quad (13.1)$$

Proof. Take a 2ε -packing. If two distinct packing signatures were assigned to the same configuration, that configuration would produce a single response within ε of both. The triangle inequality would then put the two signatures at distance at most 2ε , contradicting the packing property. $\square$

13.3 Covering construction

In the unrestricted response model, if $\text{Cov}(\Sigma_S, \varepsilon) < \infty$, the infimum is attained because covering cardinalities are integers. Choose an ε -net of cardinality $\text{Cov}(\Sigma_S, \varepsilon)$, store the index of a nearest

representative, and implement the corresponding representative response. Then every signature is approximated within ε . Consequently,

$$\text{Pack}(\Sigma_S, 2\varepsilon) \leq K^*(\varepsilon) \leq \text{Cov}(\Sigma_S, \varepsilon). \quad (13.2)$$

For a general restricted implementation architecture, the covering upper bound need not be achievable; the lower packing obstruction remains architecture-independent only within the stated state/output model.

13.4 Zero-error endpoint

For a finite metric space Σ with at least two points, let

$$\Delta(\Sigma) = \min_{x \neq y} d(x, y) > 0.$$

Whenever $0 < \varepsilon < \Delta(\Sigma)/2$, one has

$$\text{Pack}(\Sigma, 2\varepsilon) = \text{Cov}(\Sigma, \varepsilon) = |\Sigma|.$$

Thus, in the finite case, the approximate formulation exactly recovers the cardinality of the semantic quotient as the zero-error limit.

Chapter 14

Finite-Range Sparse Prime-Power Representations

14.1 The two-parameter problem

Fix the first k primes

$$p_1 = 2, p_2 = 3, p_3 = 5, \dots, p_k.$$

For an integer height bound $H \in \mathbb{N}$, $H \geq 2$, define the finite atomic dictionary

$$A_k(H) = \{p_i^a : 1 \leq i \leq k, a \geq 0, p_i^a \leq H\}, \quad (14.1)$$

with the duplicate value 1 included only once. For an integer $n \geq 1$, define

$$\lambda_{k,H}(n) = \min \left\{ r : n = \sum_{j=1}^r \varepsilon_j a_j, \varepsilon_j \in \{-1, +1\}, a_j \in A_k(H) \right\}. \quad (14.2)$$

Because $1 \in A_k(H)$, every positive integer is representable.

For an integer target bound $X \in \mathbb{N}$, $X \geq 2$, define

$$M_k(X, H) = \max_{\substack{p \leq X \\ p \text{ prime}}} \lambda_{k,H}(p), \quad (14.3)$$

and define the diagonal specialization

$$M_k(B) := M_k(B, B).$$

Keeping X and H separate is essential: an unrestricted shortest representation may use atoms larger than the target and then cancel large positive and negative terms.

14.2 Coefficient formulation

A representation can equivalently be written

$$n = \sum_{a \in A_k(H)} c_a a, \quad c_a \in \mathbb{Z},$$

with cost

$$\|c\|_1 = \sum_{a \in A_k(H)} |c_a|.$$

Thus

$$\lambda_{k,H}(n) = \min \left\{ \|c\|_1 : \sum_a c_a a = n \right\}.$$

Repeated atoms are charged with multiplicity.

14.3 Basic monotonicity

Proposition 14.1. *For fixed k, n , if $H_1 \leq H_2$, then*

$$\lambda_{k,H_2}(n) \leq \lambda_{k,H_1}(n).$$

Consequently, for fixed k, X ,

$$M_k(X, H_2) \leq M_k(X, H_1).$$

Proof. The atom dictionary increases with the height bound. □

Proposition 14.2. *For fixed X, H and $k \geq 1$,*

$$\lambda_{k+1,H}(n) \leq \lambda_{k,H}(n), \quad M_{k+1}(X, H) \leq M_k(X, H).$$

Proof. The $(k+1)$ -prime dictionary contains the k -prime dictionary. □

Remark 14.3. For a fixed target n , increasing H cannot increase $\lambda_{k,H}(n)$. The diagonal extremal quantity $M_k(B, B)$ need not be monotone in B because its target set also grows.

14.4 Counting the atomic dictionary

For fixed k ,

$$|A_k(H)| = 1 + \sum_{i=1}^k \left\lfloor \frac{\log H}{\log p_i} \right\rfloor = C_k \log H + O_k(1), \quad C_k = \sum_{i=1}^k \frac{1}{\log p_i}. \quad (14.4)$$

Hence the dictionary is logarithmically sparse in H .

14.5 Counting barrier

Proposition 14.4 (Counting lower bound). *Fix $k \geq 1$. Then, for every X, H in the finite-height model,*

$$M_k(X, H) \geq \frac{\log \pi(X)}{\log(2|A_k(H)|)} - 1. \quad (14.5)$$

Equivalently, if $M_k(X, H) \leq r$, then

$$\pi(X) \leq (2|A_k(H)|)^{r+1}. \quad (14.6)$$

Proof. There are at most $2|A_k(H)|$ signed choices per position, so the number of ordered expressions of lengths $0, 1, \dots, r$ is at most

$$1 + (2|A_k(H)|) + \dots + (2|A_k(H)|)^r \leq (2|A_k(H)|)^{r+1}.$$

If all primes $p \leq X$ had length at most r , they would all occur among these expressions. Hence

$$\pi(X) \leq (2|A_k(H)|)^{r+1}.$$

Taking logarithms gives

$$r + 1 \geq \frac{\log \pi(X)}{\log(2|A_k(H)|)},$$

and therefore

$$M_k(X, H) \geq \frac{\log \pi(X)}{\log(2|A_k(H)|)} - 1.$$

□

Corollary 14.5 (Polynomial-height regime). *If k is fixed and $H \leq X^\alpha$ for some fixed $\alpha > 0$, then*

$$M_k(X, H) \geq (1 - o(1)) \frac{\log X}{\log \log X}. \quad (14.7)$$

In particular, on the diagonal,

$$M_k(B) = M_k(B, B) \geq (1 - o(1)) \frac{\log B}{\log \log B}.$$

14.6 Deriving the lower bound with explicit quantifiers

Fix $k \geq 1$ and a height function $H = H(X) \geq 2$. If $M_k(X, H) \leq \ell$, then every prime $p \leq X$ occurs as a signed expression of length at most ℓ . The number of ordered signed expressions of length at most ℓ is at most $(2|A_k(H)|)^{\ell+1}$. Therefore

$$\pi(X) \leq (2|A_k(H)|)^{\ell+1}. \quad (14.8)$$

Taking logarithms gives

$$\ell + 1 \geq \frac{\log \pi(X)}{\log(2|A_k(H)|)}. \quad (14.9)$$

Thus

$$M_k(X, H) \geq \frac{\log \pi(X)}{\log(2|A_k(H)|)} - 1. \quad (14.10)$$

When k is fixed and $H \leq X^\alpha$ with fixed $\alpha > 0$,

$$|A_k(H)| \leq C_k \alpha \log X + O_{k,\alpha}(1) = O_{k,\alpha}(\log X).$$

Hence

$$\log(2|A_k(H)|) \leq \log \log X + O_{k,\alpha}(1).$$

Together with

$$\log \pi(X) = \log X - \log \log X + o(\log X),$$

this implies, for every fixed $\varepsilon > 0$ and all sufficiently large X ,

$$M_k(X, H) \geq (1 - \varepsilon) \frac{\log X}{\log \log X}. \quad (14.11)$$

14.7 An additive-combinatorial reformulation

Let $\mathcal{A} = A_k(H)$ and $\pm\mathcal{A} = \{\pm a : a \in \mathcal{A}\}$. Then the set of values representable with at most r atoms is

$$\mathcal{E}_{\leq r} = \bigcup_{j=0}^r j(\pm\mathcal{A}), \quad (14.12)$$

where $j(\pm\mathcal{A})$ denotes the j -fold sumset. The conjectural upper bound is consequently an additive-coverage statement about a sparse arithmetic dictionary. Counting descriptions controls possible words, but not the number of distinct integers after all additive collisions are taken into account.

14.8 Unconditional upper bounds

Because $2 \in A_k(H)$ whenever $H \geq 2$, ordinary signed-binary representations always give logarithmic upper bounds. A non-adjacent form (NAF) gives the sharper leading constant provided the height bound contains the highest power of 2 used by the NAF. A safe uniform hypothesis is $H \geq 2X$:

$$M_k(X, H) \leq \frac{\log X}{2 \log 2} + O(1) \quad (H \geq 2X). \quad (14.13)$$

Thus in the diagonal regime $H = X$ we retain the unconditional bound

$$M_k(B) = O(\log B), \quad (14.14)$$

while the sharper NAF constant is available, for example, in the enlarged-height regime $H \geq 2X$.

14.9 Proof of the logarithmic upper bounds

For completeness, every positive integer has a non-adjacent form (NAF), a signed binary expansion

$$n = \sum_{j=0}^L d_j 2^j, \quad d_j \in \{-1, 0, 1\},$$

in which no two consecutive digits are nonzero. Such an expansion can be constructed by the usual carry rule: when an odd positive integer m is encountered, choose $d_j = 1$ if $m \equiv 1 \pmod{4}$ and $d_j = -1$ if $m \equiv 3 \pmod{4}$. Then the next quotient $(m - d_j)/2$ is even, so the next digit is forced to be zero. The process terminates because this quotient has strictly smaller absolute value than m .

If the highest occupied position is L , the non-adjacency condition implies that there are at most $\lceil (L+1)/2 \rceil$ nonzero digits. The safe general bound is

$$L \leq \lfloor \log_2 n \rfloor + 1,$$

not $L \leq \log_2 n$; the latter can fail because a NAF may use a leading power larger than n , for example $3 = 4 - 1$. Consequently

$$\text{wt}_{\text{NAF}}(n) \leq \frac{\log n}{2 \log 2} + O(1). \quad (14.15)$$

Moreover $2^L \leq 2n$, so for $n \leq X$ every NAF atom is at most $2X$. Hence

$$\lambda_{k,2X}(n) \leq \frac{\log n}{2 \log 2} + O(1) \quad (14.16)$$

for every $k \geq 1$, and therefore

$$M_k(X, H) \leq \frac{\log X}{2 \log 2} + O(1) \quad (H \geq 2X). \quad (14.17)$$

For the diagonal regime $H = X$, ordinary binary expansion avoids any height issue: every power used for $n \leq X$ is at most X . Therefore

$$\lambda_{k,X}(n) \leq 1 + \lfloor \log_2 n \rfloor \quad (14.18)$$

and

$$M_k(B) \leq \log_2 B + 1 = \frac{\log B}{\log 2} + 1. \quad (14.19)$$

The argument uses only the availability of the powers 2^j in the dictionary. Additional primes can only reduce the minimum length. Thus these upper bounds are uniform in k once $k \geq 1$.

14.10 The Pulickal Conjecture

Conjecture 14.6 (Pulickal Conjecture). *For every fixed $k \geq 2$,*

$$\boxed{M_k(B) = \Theta_k\left(\frac{\log B}{\log \log B}\right)} \quad (B \rightarrow \infty).$$

The lower bound is proved; the matching upper bound is conjectural.

14.11 Why the conjecture is nontrivial

A counting argument counts formal expressions, not distinct represented integers. Pure powers form a sparse union of geometric progressions and possess substantial additive structure and congruence restrictions. The difficult direction is therefore an additive-coverage theorem for actual signed sumsets. The problem is related to established work on arithmetic diameter and representations by sums or differences of powers, power products, and multi-base Hamming weight, but those richer results do not by themselves settle the present pure-power, prime-target, finite-height conjecture.[8, 9, 10, 17]

14.12 Exact finite computation

Fix k, X, H . Let

$$E_0(k, H) = \{0\}, \quad E_{h+1}(k, H) = E_h(k, H) + (\pm A_k(H)).$$

Then E_h is exactly the set of integers representable with h signed atoms, and

$$\lambda_{k,H}(p) = \min\{h : p \in E_h(k, H)\}.$$

Every h -term sum lies in $[-hH, hH]$, so a bitset covering that full interval is an exact representation of the layer. Truncating the interval can invalidate later states when large terms cancel.

14.13 Exact computation as a finite certificate

For fixed (k, X, H) , the layer construction is a finite certificate. The dictionary $A_k(H)$ is explicit, and

$$E_0 = \{0\}, \quad E_{j+1} = E_j + (\pm A_k(H))$$

contains exactly the integers with a representation of length j . Thus first appearance of a target at layer j certifies that its minimum length is j .

A bitset implementation maps the integer interval $[-jH, jH]$ to consecutive memory positions. Every shift is by an exact integer atom, so no floating-point arithmetic is used. The full interval is required: a partial sum can be much larger than the final target and later cancel. Truncating intermediate layers to $[-X, X]$ can therefore destroy valid shortest representations.

The archived diagonal computation uses this exact layered principle. Agreement of finitely many values is finite computational evidence only; it does not prove the asymptotic conjecture.

14.14 Verified diagonal data

The archived computation for $S_4 = \{2, 3, 5, 7\}$ is diagonal, with $B = 10^2, 10^3, \dots, 10^7$. The present audit independently rechecked the six displayed record identities by exact integer arithmetic and independently reproduced the smaller portion of the diagonal table needed for the checks described here. The full 10^2 – 10^7 table is retained as archived computational data, not represented as exhaustively reverified by this audit.

B	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 6$	$k = 10$
10^2	3	3	2	2	2	2
10^3	4	4	4	4	3	3
10^4	5	5	5	5	4	4
10^5	6	6	5	5	5	5
10^6	7	7	6	6	6	5
10^7	8	7	7	6	6	6

For S_4 , the first observed record primes requiring successive lengths are

$$2, 11, 107, 1123, 11717, 607819, 7100879,$$

with lengths 1, 2, 3, 4, 5, 6, 7, respectively. The shortest representations used in the archived computation include

$$11 = -2^4 + 3^3,$$

$$107 = -2^{11} - 2^5 + 3^7,$$

$$1123 = -7^5 - 2^{10} - 3^6 + 3^9,$$

$$11717 = -2^{19} + 5^3 + 7^3 + 2^{12} + 3^{12},$$

$$607819 = -5^{10} - 2^7 + 2^4 + 5^8 + 3^{13} + 2^{23},$$

$$7100879 = -5^{10} - 7^4 - 2^7 + 2^{15} + 3^{10} + 2^{23} + 2^{23}.$$

Each identity evaluates exactly, and the repeated 2^{23} contributes two atomic terms.

Integrity note. The finite records above must be indexed by the corresponding diagonal target bound, not by the truncated strings “102”, “103”,... or “107”. The intended bounds are $10^2, 10^3, \dots, 10^7$. The separate variables (X, H) should be retained whenever finite-height and unrestricted questions are compared.

14.15 Prime targets versus all integers

Define

$$N_k(X, H) = \max_{1 \leq n \leq X} \lambda_{k,H}(n).$$

Then

$$M_k(X, H) \leq N_k(X, H).$$

The counting threshold has the same first-order logarithmic scale for primes and all integers because $\log \pi(X) = \log X - \log \log X + o(\log X)$. Whether prime targets exhibit different extremal constants or fluctuations is open.

14.16 Unrestricted representations and height inflation

Define

$$\lambda_k(n) = \min \left\{ \|c\|_1 : n = \sum_a c_a a, \ a = p_i^m, \ m \geq 0 \right\},$$

and

$$M_k^\infty(X) = \max_{\substack{p \leq X \\ p \text{ prime}}} \lambda_k(p).$$

Then

$$M_k^\infty(X) \leq M_k(X, H)$$

for every H . For each fixed n , define

$$H_k^*(n) = \min\{H \geq 2 : \lambda_{k,H}(n) = \lambda_k(n)\}.$$

This exists because an unrestricted shortest representation uses finitely many atoms. Likewise

$$H_k^*(X) = \min\{H \geq 2 : M_k(X, H) = M_k^\infty(X)\}$$

exists for every finite X , because only finitely many prime targets occur.

A natural height-control question is whether, for some fixed $\alpha_k < \infty$, every prime $p \leq X$ has an unrestricted shortest representation using atoms at most X^{α_k} for all sufficiently large X . This would connect the bounded-height and unrestricted problems.

14.17 Stabilization of unrestricted shortest representations

For each fixed target n , the unrestricted minimum $\lambda_k(n)$ is attained by a finite representation because powers of 2 alone give a representation. Choosing a minimizing representation and taking the largest atom in it gives a height H with

$$\lambda_{k,H}(n) = \lambda_k(n).$$

For a finite collection of prime targets $p \leq X$, taking the maximum of their individual stabilization heights gives $H_k^*(X)$. Hence bounded and unrestricted minima agree on every fixed finite target set once the height is sufficiently large, although this elementary existence statement gives no polynomial bound on the required height.

14.18 A direct binary upper bound

When $H \geq X$, every $n \leq X$ can be expressed by its ordinary binary expansion using only atoms at most X . Therefore

$$\lambda_{k,H}(n) \leq 1 + \lfloor \log_2 n \rfloor. \tag{14.20}$$

Signed non-adjacent representations improve the leading constant in the number of nonzero digits when the height bound is large enough to contain the possible leading NAF power (for example, $H \geq 2X$). On the diagonal $H = X$, the rigorously established unconditional statement here remains the ordinary-binary $O(\log X)$ bound.

14.19 Research directions

The main concrete questions are: determine hard primes and their profiles; compare prime and all-integer extrema; prove additive expansion for the pure-power signed sumsets; determine dependence on k and the height regime; and quantify the gap between diagonal and unrestricted shortest representations.

Warning 14.7. The Pulickal Conjecture is open. The finite computations and the counting lower bound do not prove its matching upper bound.

Chapter 15

Complex Stochastic Amplitudes: Corrected Formulation

15.1 The original construction and the normalization problem

Suppose $(\Omega, \mathcal{F}, \mu)$ is a measure space and $\psi \in L^1(\Omega, \mu) \cap L^2(\Omega, \mu)$ satisfies

$$\int_{\Omega} |\psi|^2 d\mu = 1.$$

An earlier exploratory construction proposed

$$P(A) = \left| \int_A \psi d\mu \right|^2. \quad (15.1)$$

The formula is mathematically meaningful, but it does *not* define an ordinary probability measure merely because $\|\psi\|_2 = 1$.

Indeed, if $\Omega = \{\omega_1, \omega_2\}$ with counting measure and $\psi(\omega_1) = \psi(\omega_2) = 1/\sqrt{2}$, then

$$\|\psi\|_2^2 = 1$$

but

$$P(\Omega) = \left| \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right|^2 = 2.$$

Thus countable additivity and normalization fail in general.

15.2 The correct object

Let

$$\nu(A) = \int_A \psi d\mu. \quad (15.2)$$

Then ν is a finite complex measure, and

$$Q(A) = |\nu(A)|^2 \quad (15.3)$$

is a nonnegative set function. For pairwise disjoint A, B, C , direct expansion of the squared modulus gives the grade-2 identity

$$Q(A \cup B \cup C) = Q(A \cup B) + Q(A \cup C) + Q(B \cup C) - Q(A) - Q(B) - Q(C), \quad (15.4)$$

so Q need not be additive but is a standard quantum-measure-type object.[\[12\]](#)

The right status is therefore:

A complex-amplitude model is a useful mathematical representation of interference. It is not automatically a new classical probability theory.

15.3 Stochastic phase dynamics

Let ϕ_t be a real random phase and define

$$\psi_t(\omega) = \psi_0(\omega)e^{i\phi_t(\omega)}. \quad (15.5)$$

A common model is Brownian or Gaussian phase noise. If

$$\phi_t \sim N(0, \sigma^2 t),$$

then

$$\mathbb{E}[e^{i\phi_t}] = e^{-\sigma^2 t/2}. \quad (15.6)$$

This calculation is exact.

However, ensemble “classicalization” depends on cross-phase correlations. The central object is not only $\mathbb{E}[e^{i\phi_t(\omega)}]$ but

$$K_t(\omega, \omega') = \mathbb{E}[e^{i(\phi_t(\omega) - \phi_t(\omega'))}]. \quad (15.7)$$

A meaningful research problem is therefore to determine conditions on K_t under which the expected interference cross-terms vanish.

15.4 Interference expansion and the role of cross terms

For pairwise disjoint measurable sets $A_1, \dots, A_m$,

$$\nu\left(\bigcup_j A_j\right) = \sum_j \nu(A_j).$$

Consequently

$$Q\left(\bigcup_j A_j\right) = \sum_j |\nu(A_j)|^2 + 2\Re \sum_{j < \ell} \nu(A_j) \overline{\nu(A_\ell)}. \quad (15.8)$$

The cross terms are the exact mathematical location of interference. Their behavior, rather than the one-point phase mean alone, determines whether an averaged model becomes approximately additive.

15.5 Measurability conditions for the random model

For a fully specified stochastic model, place the phases on a probability space $(\Xi, \mathcal{G}, \mathbb{P})$ and regard $\phi_t(\omega, \xi)$ as jointly measurable. Since $|e^{i\phi_t}| = 1$, the condition $\psi_0 \in L^1(\Omega, \mu)$ implies

$$|\psi_0(\omega)e^{i\phi_t(\omega, \xi)}| = |\psi_0(\omega)|,$$

so each realization remains L^1 whenever measurability holds. Further exchanges of integration and expectation should then be justified by the appropriate Tonelli/Fubini or dominated-convergence hypotheses.

15.6 A corrected research model

Define

$$\nu_t(A) = \int_A \psi_0(\omega) e^{i\phi_t(\omega)} d\mu(\omega), \quad Q_t(A) = |\nu_t(A)|^2. \quad (15.9)$$

Questions that remain legitimate are:

1. When does $\mathbb{E}Q_t(A)$ approach an additive measure?
2. How fast do interference terms decay?
3. Which correlation kernels K_t give complete decoherence versus partial coherence?

This is a strong research direction precisely because it no longer claims more than the mathematics provides.

Chapter 16

The Three-Outcome Coin Model

16.1 Definition

A physical coin can in principle land heads, tails, or edge; finite-thickness coin dynamics have been studied experimentally and analytically, including explicit edge probabilities.[15] Let $N \in [0, 1]$ be the edge probability. A symmetric model is

$$P(H) = P(T) = \frac{1-N}{2}, \quad P(E) = N. \quad (16.1)$$

This is an ordinary three-outcome probability model.

16.2 Entropy

Its Shannon entropy is

$$H(N) = -\frac{1-N}{2} \log \frac{1-N}{2} - \frac{1-N}{2} \log \frac{1-N}{2} - N \log N, \quad (16.2)$$

so

$$H(N) = -(1-N) \log \frac{1-N}{2} - N \log N.$$

(16.3)

At the endpoints,

$$H(0) = \log 2, \quad H(1) = 0,$$

and the maximum occurs when the three outcomes are equally likely:

$$N = \frac{1}{3}, \quad H = \log 3. \quad (16.4)$$

The value $N = 1/2$ is not a phase transition. It merely makes the edge probability equal to the sum of the two symmetric face probabilities:

$$N = P(H) + P(T).$$

16.3 Entropy derivation

With face probabilities $(1-N)/2$ and edge probability N ,

$$H(N) = -2 \frac{1-N}{2} \log \left(\frac{1-N}{2} \right) - N \log N. \quad (16.5)$$

For $0 < N < 1$,

$$H'(N) = \log\left(\frac{1-N}{2N}\right), \quad H''(N) = -\frac{1}{1-N} - \frac{1}{N} < 0. \quad (16.6)$$

Thus $N = 1/3$ is the unique global maximizer, with $H = \log 3$. The endpoint values are $H(0) = \log 2$ and $H(1) = 0$ under the usual convention $0 \log 0 = 0$.

16.4 Why the corrected model is useful

The model is useful as a clean reminder that “two outcomes” need not be a universal abstraction for an approximately physical coin experiment. But nothing here creates a new probability calculus. It is best viewed as an elementary noisy-channel or mixture model.

Chapter 17

A Real Surrogate from the Lambert W Function

17.1 The problem

Suppose one wishes to assign a positive real surrogate to the imaginary unit by a fixed scalar criterion rather than by an alleged algebraic replacement. One natural construction is to match the magnitude of

$$f(z) = \frac{\operatorname{Log} z}{z}$$

at $z = i$.

Since

$$\operatorname{Log} i = \frac{\pi}{2}i, \quad \frac{\operatorname{Log} i}{i} = \frac{\pi}{2},$$

the target magnitude is $\pi/2$.

We therefore ask for the positive x satisfying

$$\left| \frac{\log x}{x} \right| = \frac{\pi}{2}. \quad (17.1)$$

17.2 Uniqueness on $(0, 1)$

For $0 < x < 1$, equation (17.1) is

$$-\frac{\log x}{x} = \frac{\pi}{2},$$

or

$$\log x = -\frac{\pi}{2}x.$$

Exponentiating gives

$$x = e^{-\pi x/2}.$$

Set $y = \pi x/2$. Then

$$\frac{2y}{\pi} = e^{-y}, \quad ye^y = \frac{\pi}{2}.$$

Thus

$$y = W_0\left(\frac{\pi}{2}\right), \quad (17.2)$$

and

$$x_* = \frac{2}{\pi} W_0\left(\frac{\pi}{2}\right). \quad (17.3)$$

Numerically this is approximately

$$x_* \approx 0.474541.$$

17.3 A direct monotonicity check

On $(0, 1)$ the function $g(x) = -\log x/x$ is strictly decreasing from $+\infty$ to 0. On $[1, \infty)$ the function $h(x) = \log x/x$ has its maximum $1/e$, since $h'(x) = (1 - \log x)/x^2$. Because $1/e < \pi/2$, the magnitude equation has no solution for $x \geq 1$. Thus the positive solution is unique before the Lambert W inversion is invoked.

Theorem 17.1. *There is exactly one positive solution to (17.1), namely*

$$x_* = \frac{2}{\pi} W_0(\pi/2).$$

Proof. For $0 < x < 1$, the function $g(x) = -\log x/x$ is strictly decreasing because

$$g'(x) = \frac{\log x - 1}{x^2} < 0.$$

Moreover $g(x) \rightarrow \infty$ as $x \downarrow 0$ and $g(x) \rightarrow 0$ as $x \uparrow 1$, so there is exactly one solution in $(0, 1)$. For $x \geq 1$, one has $|\log x|/x = \log x/x \leq 1/e < \pi/2$, so there is no solution there. The Lambert W derivation identifies the unique positive solution. $\square$

17.4 Interpretation

The number x_* is the unique positive scalar selected by the stated magnitude-matching criterion. It is not a new algebraic image of i , not a replacement satisfying $x_*^2 = -1$, and not a contradiction of the complex numbers.

This is an excellent example of how an initially speculative phrase can be converted into a precise theorem by narrowing the claim.

Chapter 18

Complex Vector Arithmetic and Realification

18.1 The representation

Write

$$z = a + bi \quad (18.1)$$

and represent it by the real vector

$$\rho(z) = \begin{pmatrix} a \\ b \end{pmatrix}. \quad (18.2)$$

Multiplication by $u + vi$ is represented by

$$M(u, v) = \begin{pmatrix} u & -v \\ v & u \end{pmatrix}. \quad (18.3)$$

Then

$$\rho((u + vi)(a + bi)) = M(u, v)\rho(a + bi).$$

18.2 Inner product and norm

The ordinary Hermitian inner product on $\mathbb{C}^n$ can be represented by real block matrices. In particular, for $z = a + bi$,

$$|z|^2 = a^2 + b^2.$$

There is nothing pathological here: the complex vector space is simply being viewed as a real vector space of twice the dimension.

Warning 18.1. A norm is positive definite, not merely semidefinite. Any formulation that calls the standard modulus or Euclidean norm “semidefinite” is weaker than necessary.

18.3 Worked RLC example

For an AC circuit with

$$R = 10\,\Omega, \quad L = 0.02\,\text{H}, \quad C = 500\,\mu\text{F}, \quad \omega = 100\,\text{rad/s},$$

the series impedance is

$$Z = R + i \left(\omega L - \frac{1}{\omega C} \right) = 10 - 18i. \quad (18.4)$$

With phasor voltage $V = 10$, the current is

$$I = \frac{10}{10 - 18i} = 0.235849 + 0.424528i. \quad (18.5)$$

The corresponding real signal is

$$i(t) = 0.235849 \cos(100t) - 0.424528 \sin(100t) \quad (18.6)$$

under the usual phasor convention.

The matrix representation of multiplication by $10 - 18i$ is

$$\begin{pmatrix} 10 & 18 \\ -18 & 10 \end{pmatrix}. \quad (18.7)$$

The magnitude and phase can also be read directly from the impedance:

$$|Z| = \sqrt{10^2 + 18^2} = \sqrt{424}, \quad \arg Z = -\arctan(18/10).$$

Thus the current phasor has the opposite phase shift. Taking the real part of $Ie^{i\omega t}$ yields the displayed time-domain signal under the stated convention.

The calculation is correct, useful, and conventional. The novelty claim should be modest: it is an explicit pedagogical realification of complex arithmetic.

Chapter 19

A Four-Valued NAND Algebra

19.1 The proposed finite carrier

Let

$$V = \{-1, 0, 1, i\}.$$

Consider the algebra with carrier V and one binary fundamental operation $N : V^2 \rightarrow V$ given by the table; no additional constant symbols are assumed in the term language. The operation is not asserted to be a complete logical basis.

N	-1	0	1	i
-1	1	1	1	1
0	1	0	0	0
1	1	0	-1	-1
i	1	0	-1	i

This table is mathematically legitimate because a finite algebraic operation is determined by its table.

19.2 What the table does not prove

An earlier draft claimed that this NAND is functionally complete. That claim is false.

Proposition 19.1. *The NAND operation above is not functionally complete on V .*

Proof. Consider the subset $\{0, i\} \subset V$. From the table,

$$N(0, 0) = 0, \quad N(0, i) = 0, \quad N(i, 0) = 0, \quad N(i, i) = i.$$

Therefore any expression built from variables whose values lie in $\{0, i\}$ remains in $\{0, i\}$. Such an expression can never produce 1 or -1 . Hence the clone generated by N cannot contain all functions $V^n \rightarrow V$. $\square$

19.3 The research question that remains

The right object is not “imaginary universal logic” but the algebra

$$\mathfrak{A} = (V, N).$$

One can ask for its term functions, subalgebras, automorphisms, congruences, and expressive power. A second operation could be added and functional completeness revisited only after a proof. The word “NAND” here is a name for the displayed finite binary operation. Classical Boolean NAND is a connective on a two-valued truth set; a four-valued algebra additionally requires a semantic interpretation, designated values, and a precise definition of functional completeness.

A second operation could be added and functional completeness revisited only after a proof.

This is a nice example of a construction surviving after an overstrong claim is removed.

Chapter 20

The Sleeping Beauty Problem: A Sample-Space Lesson

20.1 The original answer

The archived paper argued for

$$P(H \mid \text{awake}) = \frac{1}{3}, \quad P(T \mid \text{awake}) = \frac{2}{3}.$$

The conclusion is compatible with the standard thirder interpretation, but the original Bayesian derivation was not valid because it wrote

$$P(W \mid H) = \frac{1}{3}, \quad P(W \mid T) = \frac{2}{3}$$

by counting centered awakenings. Those are not ordinary conditional probabilities on the experiment unless the sample space has been enlarged to include a random awakening index.

20.2 One coherent centered sampling model

Define the centered-world space

$$\Omega_c = \{(H, M), (T, M), (T, T)\},$$

where the second coordinate denotes the awakening center. Assign equal probability $1/3$ to these three centered worlds. Then

$$P_c(H) = \frac{1}{3}, \quad P_c(T) = \frac{2}{3}.$$

This is a coherent model because it explicitly changes the sample space from “coin toss” to “coin toss plus awakening index.”

20.3 The halfer alternative

A different question is what Beauty should believe while preserving the original trial-centered sample space and conditioning only on information that does not identify a random awakening index. That approach can maintain probability $1/2$ for the coin.

Under the centered sampling model, conditioning is ordinary because the awakening center has been included in the sample point. Under the trial-centered model, the original random experiment is only the coin toss. The numerical disagreement therefore reflects different probability models for self-locating information, not an algebraic disagreement about Bayes' theorem.

The philosophical dispute therefore cannot be settled by writing Bayes' theorem alone. One must specify the representation of self-locating evidence and the sampling protocol.

Status: Lesson retained. The valuable insight is not “Bayes proves one side.” It is that self-location changes the probabilistic model, and the sample space must be stated explicitly.

Chapter 21

Complex-Coefficient Oscillators and the Meaning of Damping

Consider

$$mx'' + ix' + kx = 0. \quad (21.1)$$

The equation is mathematically legitimate: it is a second-order linear ODE with complex coefficients. Looking for $x(t) = e^{\lambda t}$ gives

$$m\lambda^2 + i\lambda + k = 0.$$

The phrase “imaginary damping” must, however, be treated carefully. In a standard real damped oscillator, a term proportional to x' contributes a real damping coefficient and typically creates roots with negative real part. Here the coefficient is i , so the qualitative behavior is governed by the discriminant

$$(-i)^2 - 4mk = -1 - 4mk.$$

For positive m, k , the roots are purely imaginary. The system is therefore oscillatory rather than conventionally dissipative.

The safe language is:

This is a complex-coefficient oscillator. Calling the ix' term “damping” is a physical interpretation that requires additional modeling assumptions; it is not a consequence of the equation alone.

Chapter 22

The Failed Recursive-Imaginary Programme: What Survives

22.1 Recursive imaginary units

The exploratory definition

$$\mathbf{i}_0 = 1, \quad \mathbf{i}_1 = \mathbf{i}, \quad \mathbf{i}_{n+1} = \sqrt{-\mathbf{i}_n} \quad (22.1)$$

can be made mathematically exact once the principal square root is fixed.

If

$$\mathbf{i}_n = e^{i\theta_n}, \quad \theta_n \in (-\pi, \pi],$$

then

$$\theta_{n+1} = \frac{1}{2} \text{PV}(\theta_n + \pi) \quad (22.2)$$

under the corresponding principal conventions. There is no need to treat $\mathbf{i}_n$ as a new independent number system.

22.2 What fails

Several former interpretations are not retained.

1. The symbols $\mathbf{i}_n$ are ordinary complex numbers and do not form an independent basis of a higher-dimensional algebra merely by being labeled differently.
2. A recursively defined “conjugate” that sends $e^{i\theta_n}$ to $e^{-i\theta_n}$ is just ordinary complex conjugation.
3. Products of unit-modulus recursive roots do not decay in magnitude merely because their phases vary.
4. A recursively defined Jacobian of the form $c_N J$ with scalar $c_N = \sum \mathbf{i}_n$ is still an ordinary scalar multiple of J .
5. A principal-root Laplace transform evaluated at a recursive complex value is still an ordinary Laplace transform at a complex parameter.
6. A Mellin transform with a shifted exponent is still the ordinary Mellin transform with a shifted complex argument.

22.3 The productive replacement

The recursive-root idea is worth preserving as a study of branch-sensitive contractions, cycles, and inverse geometry. It need not become a new ontology of imaginary numbers.

Historical note. The strongest version of the recursive-root work is therefore the most classical version: exact branch conventions, exact recurrences, explicit inverse levels, and careful limits.

Chapter 23

Transform Reductions

23.1 Imaginary Laplace transform

If $J \in \mathbb{C}$ is chosen as the transform parameter, then the most economical notation is simply

$$\mathcal{L}\{f\}(J) = \int_0^\infty f(t)e^{-Jt} dt, \quad (23.1)$$

provided the integral converges. There is no remaining free variable s in this definition. If one wants to emphasize that a particular complex number has been selected, one may write $\mathcal{L}_J[f]$ for the scalar value

$$\mathcal{L}_J[f] := \mathcal{L}\{f\}(J), \quad (23.2)$$

but then the subscript J is the parameter being evaluated at, not an additional transform variable.

23.2 Laplace convergence region

For $J = \sigma + i\tau$,

$$|e^{-Jt}| = e^{-\sigma t}.$$

Thus $\Re J$ controls decay or growth, while $\Im J$ contributes oscillation only. This is the ordinary analytic structure of the complex Laplace transform. A purely imaginary evaluation parameter is therefore a boundary-type oscillatory case, and convergence has to be checked from the particular function f .

23.3 Imaginary Mellin transform

If one defines

$$\mathcal{M}_J\{f\}(s) = \int_0^\infty f(t)t^{s+J-2} dt, \quad (23.3)$$

then this is the ordinary Mellin transform at a shifted argument:

$$\mathcal{M}_J\{f\}(s) = \mathcal{M}\{f\}(s + J - 1). \quad (23.4)$$

For $c > 0$,

$$\mathcal{M}_J\{f(ct)\}(s) = c^{-(s+J-1)} \mathcal{M}_J\{f\}(s). \quad (23.5)$$

The correction matters: any scaling law that omits the dependence on s is dimensionally incomplete.

23.4 Mellin convergence and the shifted argument

The substitution $u = ct$ with $c > 0$ gives

$$t^{s+J-2} = c^{-(s+J-2)} u^{s+J-2}, \quad dt = du/c,$$

so the total scaling factor is

$$c^{-(s+J-1)}.$$

The shift in the Mellin argument is exactly $s \mapsto s + J - 1$. Convergence near 0 and ∞ is governed by the real part of this shifted parameter together with the asymptotic behavior of f . Thus any claimed “imaginary Mellin” phenomenon should begin with an ordinary Mellin convergence strip and then ask what happens when the argument is moved vertically in the complex plane.

23.5 Fourier caution

The whole-line Fourier example $f(t) = e^{-2t}$ is not integrable on $\mathbb{R}$ because it diverges as $t \rightarrow -\infty$. Parseval or orthogonality statements require the actual hypotheses of the Fourier-space theorem being invoked.

The general lesson is useful: an integral transform should be checked first for existence and only then extended or reinterpreted.

Chapter 24

Recursive Probability, Entropy, and Measures: What Survives

24.1 Complex weights are not classical probabilities

A complex weight can be useful in wave or interference models, but a quantity such as

$$\left| \int_A \psi \, d\mu \right|^2$$

should not be called an ordinary probability measure without proof of additivity and normalization.

24.2 Quantum-measure interpretation

The amplitude-squared construction is naturally related to quantum measure theory. In particular, if ν is a complex measure then

$$Q(A) = |\nu(A)|^2$$

is nonnegative but not generally additive. Its structure is weaker than classical measure theory and stronger than an arbitrary set function.

24.3 A useful normalized finite-dimensional variant

Let $\mathcal{H}$ be a Hilbert space, let ψ be a unit vector, and let $\{P_j\}$ be a family of orthogonal projections summing to the identity. Then

$$\mathbb{P}(j) = \langle \psi, P_j \psi \rangle$$

is a genuine probability distribution. Complex amplitudes enter through the vector and the inner product, while the final event probabilities satisfy the required normalization because the projectors form a resolution of identity.

This formulation is standard and provides a clean baseline for any future interference model.

Chapter 25

Formal Hyper-Imaginary Spaces

25.1 The formal coordinate space

For a fixed finite N , one may define

$$\mathcal{H}_N = \mathbb{R}^{N+1}$$

with basis symbols

$$e_0, e_1, \dots, e_N.$$

One may then write formal vectors

$$v = \sum_{j=0}^N a_j e_j.$$

This is mathematically sound as a vector space.

25.2 What cannot be inferred

If one subsequently sets $e_j = i_j$ for recursively generated ordinary complex numbers, the symbols cease to be linearly independent in the literal complex plane. There is only a two-dimensional real vector space behind the complex numbers. Thus the formal coordinate model and the complex-number realization must be distinguished.

Likewise, a multiplication law on $\mathcal{H}_N$ cannot be assumed merely from a list of named basis vectors. One must explicitly define the multiplication table and then check associativity, distributivity, identity, and closure.

25.3 Linear dependence in the literal complex realization

If the recursively generated values are interpreted as ordinary complex numbers, then

$$i_j = e^{i\theta_j} = \cos \theta_j + i \sin \theta_j.$$

Every such value lies in the same two-dimensional real vector space $\text{span}_{\mathbb{R}}\{1, i\}$. The formal space $\mathbb{R}^{N+1}$ is therefore a different object: its basis vectors are symbols encoding coordinates, not ordinary complex numbers. Any algebra built on those coordinates needs an explicitly defined multiplication law rather than an informal identification of each basis vector with a complex value.

25.4 Correct perspective

The idea is useful as an abstract state-space construction, especially if the coordinates encode different recursive levels or features. It is not automatically a new number field or algebra.

Chapter 26

Lossless Compression, Topology, and Other Overstatements

Several early phrases can be repaired by replacing universal words with task-dependent ones.

26.1 Topological compression

A topological invariant is generally many-to-one. Therefore topology alone cannot guarantee lossless recovery of an original data object. What can be claimed is that an invariant may act as a task-dependent compressed representation for questions that depend only on that invariant.

26.2 Approximate Yoneda language

The Yoneda lemma is a precise categorical statement about representable functors and natural transformations. A coordinate matching or similarity rule is not automatically an “approximate Yoneda lemma.” It is better to describe such constructions as approximate embeddings, finite sketches, or behavioral summaries unless the categorical types are explicitly specified.

26.3 Recursive Hodge operators

A weighted sum such as

$$\mathcal{H}_N = \sum_{n=0}^N c_n H_n$$

is mathematically meaningful if H_n are operators on a common vector space. But unless the H_n belong to genuinely different spaces or encode additional structure, the construction is simply an operator combination. Novelty has to come from the operator family and its theorem, not from the adjective “recursive.”

Chapter 27

A Clean Mathematical Inventory of the Notebook

This chapter separates retained results by logical strength.

27.1 Theorems established in the core

1. Exact radial law $|z_n| = |z_0|^{1/r^n}$.
2. Exact principal angular recurrence $\alpha_{n+1} = (\pi - \alpha_n)/r$.
3. Exact angular solution (2.14).
4. Unique nonzero attracting two-cycle $\{e^{\pm i\pi/(r+1)}\}$.
5. No nonzero periodic orbit of period greater than two for the principal negative-root map.
6. Exact inverse level as a rotated regular r^n -gon.
7. Exact Hausdorff distance and first-order asymptotic to the unit circle.
8. Uniform branch measures converge weakly to Haar measure.
9. Entropy-dimension formula for the standard self-similar branch measure under the stated separation setting.
10. Polynomial bridge: $J(-z^r) = S^1$.
11. Exact state-separation theorem in the task-semantic framework.
12. Binary memory lower bound $b \geq \lceil \log_2 D(S) \rceil$.
13. Architecture-relative capacity impossibility.
14. Packing lower bound and covering upper bound for approximate realization.
15. Counting lower bound for $M_k(X, H)$ and hence the diagonal $M_k(B)$.
16. Ordinary-binary $O(\log B)$ upper bound for the diagonal $M_k(B)$, together with the sharper NAF bound when $H \geq 2X$.
17. Existence and uniqueness of the Lambert- W surrogate.

27.2 Precise conjectures

The principal conjectural statement is

$$M_k(B) = \Theta_k \left(\frac{\log B}{\log \log B} \right)$$

for fixed $k \geq 2$.

Other open problems concern the phase-deformed branch dynamics, hard-prime statistics, approximate realization constants, and stochastic phase kernels.

27.3 Exploratory but useful objects

These include the complex-amplitude set function, the three-outcome coin model, the four-valued NAND algebra, formal hyper-imaginary coordinate spaces, and the complex-vector realification viewpoint.

27.4 Discarded claims

The following are not carried into the final theory:

1. recursive imaginary units as a genuinely new number system;
2. a universal deterministic replacement for stochasticity;
3. physical laws derived solely from the recursive-root formalism;
4. guaranteed lightning harvesting, medical prediction, financial prediction, or universal engineering performance claims unsupported by experimental models;
5. lossless topological compression as a general theorem;
6. functional completeness of the four-valued NAND table;
7. the claim that ordinary Laplace or Mellin transforms become fundamentally new transforms merely by a recursive complex parameter;
8. any global Julia-set theorem attached directly to a discontinuous principal-root branch map.

Chapter 28

The Convergence Map: How the Pieces Fit

The notebook can be summarized by the following chain:

$$\begin{array}{l} \text{recursion} \longrightarrow \text{branching} \longrightarrow \text{representation}, \\ \text{representation} \longrightarrow \text{asymptotic structure} \longrightarrow \text{effective description}. \end{array} \quad (28.1)$$

The recursive-root chapters make this literal: branch selection produces a deterministic map; retaining all branches produces an inverse tree; weighting branches produces a measure.

The Pulickal Framework makes the representation step explicit: a state should preserve exactly those distinctions required by future tasks.

The sparse representation chapter turns the same intuition into an extremal number-theoretic question: how many atomic pieces are really needed to represent a target?

The satellite chapters test what happens when the same instinct is pushed into probability, logic, transforms, and physical models. In each case the durable contribution comes from making the representation explicit and dropping the claim that was larger than the mathematics.

Thus the notebook does have a coherent theme, but it is a methodological theme rather than a claim of a single grand unified theory.

28.1 The methodological principle

Define the object; separate branch choices; identify the equivalence relation; compute the asymptotic object; then state exactly what the result does not imply.

That sentence is perhaps the closest thing this notebook has to a single theorem-shaped philosophy.

Chapter 29

Research Programme: What Is Worth Doing Next

29.1 Recursive-root dynamics

The strongest direct continuation is the phase family

$$T_{r,\beta}(\theta) = \frac{\text{PV}(\theta + \beta)}{r}.$$

Questions include:

1. exact classification of periodic branch itineraries;
2. parameter intervals supporting a given itinerary;
3. endpoint phenomena under different principal-angle conventions;
4. statistical behavior under random branch selection;
5. variable root orders r_n and exact geometric asymptotics.

29.2 Operational semantics

The next serious extension is stochastic or nondeterministic behavior. A natural replacement for a single signature is a set-valued, kernel-valued, or distribution-valued response. Approximate and probabilistic semantics should be developed only after deciding what notion of behavioral equivalence is operationally meaningful.

29.3 Sparse prime-power representations

The most ambitious open mathematical question is the Pulickal Conjecture. A realistic programme is:

1. push exact finite computation much further;
2. compare all-integer and prime-only maxima;
3. search for congruence obstructions;
4. study density-one coverage before worst-case coverage;
5. compare the pure-power dictionary with the richer mixed-power dictionary.

29.4 Complex-amplitude noise

A mathematically focused version of the stochastic-amplitude problem would study

$$K_t(\omega, \omega') = \mathbb{E}[e^{i(\phi_t(\omega) - \phi_t(\omega'))}]$$

and derive exact sufficient conditions for interference terms to vanish in the long-time limit.

29.5 Finite-valued logic

The four-valued NAND algebra is small enough for exhaustive algebraic analysis. One can compute all term operations generated by N , identify subalgebras, and determine exactly which functions are expressible.

Chapter 30

How to Extend This Notebook Without Breaking It

The following rules are intended as permanent editorial constraints.

1. Never promote a heuristic to a theorem because a numerical experiment looks clean.
2. Never call a representation “new” merely because it uses a new symbol.
3. Never identify two limiting procedures without specifying the topology or averaging operation.
4. Never turn an architecture-specific lower bound into a universal physical law.
5. Never use a probabilistic conditional probability that actually belongs to a centered or self-locating sample space without saying so.
6. Never use a branch-sensitive complex identity without tracking the branch defect.
7. When a claim fails, preserve the exact counterexample or corrected theorem whenever it is educational.
8. Stop a proof when the stated theorem is established. A proof need not explain all possible interpretations of the universe to be rigorous.

30.1 A stopping criterion against infinite regress

A chapter is complete when:

1. its objects are defined;
2. its assumptions are stated;
3. its principal theorem or claim is proved or labeled conjectural;
4. all major counterexamples within the chosen scope are addressed;
5. the boundary of the model is explicit.

After that, deeper questions are optional extensions, not hidden defects in the previous argument.

Chapter 31

Closing Perspective

There is a useful kind of mathematical maturity in taking a construction that once looked revolutionary and asking a narrower question:

What is the strongest thing here that is actually true?

For the recursive-root work, the answer is stronger than a numerical curiosity. The principal branch admits an exact contraction mechanism and a unique nonzero two-cycle; the full inverse system produces exact regular polygons whose mesh and radial errors can be quantified; the associated branch histories produce a natural symbolic tree and self-similar measures. These results belong in a serious notebook.

For the operational framework, the answer is similarly clean. A task family induces a quotient of information states, and any exact state-based implementation must separate its classes. That theorem is elementary, but elementary theorems become useful when their scope is made exact.

For the sparse prime-power problem, the answer is a precise conjecture with a proved lower barrier, a proved weaker upper bound, and exact finite computations. That is a legitimate research problem even though its central asymptotic statement remains open.

For the exploratory probability, logic, transform, and surrogate chapters, the answer is often a corrected construction rather than a new theory. That is not a failure of the notebook. It is part of the notebook's purpose.

The permanent theme is therefore not “everything reduces to recursive imaginaries.” It is narrower and more durable:

$$\begin{aligned} &\text{well-defined representation} + \text{explicit branch or equivalence structure} \\ &+ \text{correct limiting notion} \implies \text{mathematics that can survive scrutiny.} \end{aligned} \tag{31.1}$$

That is the standard to which future pages should be held.

Appendix A

Detailed Rederivation of the Principal-Root Theorem

This appendix is deliberately repetitive. It records the recursive-root argument from first principles so that the result can be audited independently of the narrative development.

A.1 From the principal logarithm to the explicit map

Let $z = \rho e^{i\theta}$ with $\rho > 0$ and $\theta \in (-\pi, \pi]$. There is a unique $\tilde{\theta} \in (-\pi, \pi]$ congruent to $\theta + \pi$ modulo 2π . Hence

$$F_r(z) = \rho^{1/r} e^{i\tilde{\theta}/r}.$$

This immediately separates modulus from argument. No hidden approximation is used.

A.2 Inductive proof of the radial formula

The one-step identity gives $|z_{n+1}| = |z_n|^{1/r}$. If $|z_n| = |z_0|^{1/r^n}$, then $|z_{n+1}| = |z_0|^{1/r^{n+1}}$. Induction proves the formula for all n .

A.3 Inductive proof of the angular-magnitude recurrence

If $\theta_n \leq 0$, then $\theta_n + \pi \in [0, \pi]$ and

$$|\theta_{n+1}| = \frac{\pi - |\theta_n|}{r}.$$

If $\theta_n > 0$, then $\theta_n - \pi \in (-\pi, 0)$ and the same absolute-value formula follows. Thus

$$\alpha_{n+1} = \frac{\pi - \alpha_n}{r}.$$

A.4 Periodic points

For a nonzero periodic point of period p ,

$$|z| = |z|^{1/r^p},$$

so $|z| = 1$. The angular magnitude is a periodic point of the affine contraction $A_r(\alpha) = (\pi - \alpha)/r$. Since A_r^p has slope $(-1/r)^p$, it has exactly one fixed point, namely $\pi/(r+1)$. The sign changes at each step, producing exactly the two-cycle $e^{\pm i\pi/(r+1)}$.

A.5 The exceptional origin

The point 0 is a fixed point because $F_r(0) = 0$. It is deliberately excluded from the nonzero periodic-orbit theorem. This is the only periodic behavior outside the unit-circle two-cycle.

Appendix B

Branch-Boundary Sanity Checks

B.1 Approach to the positive real axis

Let $x > 0$ and consider $z = x \pm i\eta$ with $\eta \downarrow 0$. Then $-z = -x \mp i\eta$. The principal arguments approach $\mp\pi$, so

$$\lim_{\eta \downarrow 0} F_r(x + i\eta) = x^{1/r} e^{-i\pi/r}, \quad \lim_{\eta \downarrow 0} F_r(x - i\eta) = x^{1/r} e^{i\pi/r}.$$

These are distinct for every $r \geq 2$, proving the discontinuity across $(0, \infty)$ directly. At the origin, however,

$$|F_r(z)| = |z|^{1/r} \rightarrow 0,$$

so the separately defined value $F_r(0) = 0$ is continuous there.

B.2 Why the principal-value convention matters

The two one-sided limits above explain why replacing the principal argument by an unrestricted real argument produces the wrong global dynamics. Any lifted formula implicitly chooses one side of the branch cut and thereby suppresses the jump. The principal map restores the jump through the principal-value representative.

Appendix C

Detailed Hausdorff Audit

C.1 First Hausdorff half-distance

For a unit point $e^{i\theta}$ and a polygon vertex $\rho e^{i\vartheta}$,

$$|e^{i\theta} - \rho e^{i\vartheta}|^2 = 1 + \rho^2 - 2\rho \cos(\theta - \vartheta).$$

The nearest vertex has angular error at most π/N , and the right-hand side is increasing in the absolute angular error on that interval. Hence the maximum is attained at an arc midpoint and equals

$$\sqrt{1 + \rho^2 - 2\rho \cos(\pi/N)}.$$

C.2 Second Hausdorff half-distance

A vertex $\rho e^{i\vartheta}$ has distance $|1 - \rho|$ from S^1 . Since

$$1 + \rho^2 - 2\rho \cos(\pi/N) = (1 - \rho)^2 + 2\rho(1 - \cos(\pi/N)),$$

the first half-distance is larger whenever $\rho > 0$. This proves the exact Hausdorff formula.

C.3 Asymptotic expansion

Let $a = \log |z_0|$ and $N = r^n$. Then

$$\rho = e^{a/N} = 1 + \frac{a}{N} + \frac{a^2}{2N^2} + O(N^{-3}),$$

and

$$\cos(\pi/N) = 1 - \frac{\pi^2}{2N^2} + O(N^{-4}).$$

Substitution gives

$$d_H^2 = \frac{a^2 + \pi^2}{N^2} + O(N^{-3}),$$

and therefore

$$d_H = \frac{\sqrt{a^2 + \pi^2}}{N} + O(N^{-2}).$$

Appendix D

Audit of the Symbolic-Measure Construction

D.1 Cylinder probabilities

For $\Sigma_r = \{0, \dots, r-1\}^{\mathbb{N}}$ and a word $w = (k_1, \dots, k_n)$, the cylinder

$$[w] = \{\omega : \omega_j = k_j, 1 \leq j \leq n\}$$

has Bernoulli mass

$$\nu_p([w]) = p_{k_1} \cdots p_{k_n}.$$

The logarithmic mass per symbol approaches $-H(p)$ for typical sequences, while the angular coding scale is r^{-n} . The entropy-over-log-scale quotient is therefore the natural dimension quantity.

D.2 Support dimension versus measure dimension

If all $p_k > 0$, the support of the Bernoulli pushforward is all of S^1 , whose Hausdorff dimension is 1. If some weights vanish, the support is the corresponding symbolic coding attractor. The measure itself can have dimension $H(p)/\log r < 1$. These are distinct statements and should never be conflated.

Appendix E

Audit of the Prime-Power Computation

E.1 Exact layer identity

For

$$E_0 = \{0\}, \quad E_{h+1} = E_h + (\pm A_k(H)),$$

every element of E_h has a representation using exactly h atoms, and every such representation produces an element of E_h . This follows by induction on h . Hence first membership in the layers is equivalent to the minimum signed atomic length.

E.2 Why full cancellation ranges are required

Although the target may satisfy $p \leq X$, a shortest representation can pass through partial sums larger than X . Since an h -term representation uses atoms at most H , every intermediate partial sum lies in $[-hH, hH]$, which is a safe exact state range. Shrinking the range without a proof of safety can lose valid representations.

E.3 Finite computation versus asymptotic proof

An exact computation certifies finitely many targets and heights. It cannot, by itself, establish an asymptotic formula as $X \rightarrow \infty$. This is why the displayed record data and the Pulickal Conjecture have different logical statuses.

Appendix F

Audit of the Operational and Approximate Frameworks

F.1 Exact quotient

The signature map σ_S identifies exactly those states that agree on every admissible task. The quotient $X/\sim_S$ is therefore the canonical set of behavioral classes. Any exact representation must separate distinct classes.

F.2 Approximate quotienting

Once a metric is put on the signature space, approximation becomes a metric quantization problem. A 2ε -packing is an obstruction because one representative cannot lie within ε of two points more than 2ε apart. An ε -cover is sufficient when arbitrary representative responses are allowed.

Appendix G

Failure-Mode Catalogue

G.1 Branch errors

Principal logarithms are not algebraic homomorphisms; missing a $2\pi i$ defect changes the result.

G.2 Domain errors

Transform identities are only meaningful where the defining integral or series converges.

G.3 Normalization errors

An L^2 -normalized amplitude does not automatically define a classical probability measure after squaring a complex integral.

G.4 Semantic errors

A longer description is not automatically a larger persistent state space. The future task family must be fixed before counting distinctions.

G.5 Novelty errors

A new name or notation is not a theorem. The mathematical content lies in the definitions, proofs, examples, and consequences.

Appendix H

Audit of the Principal Negative-Root Recursion

This appendix records the minimum calculations needed to independently rederive the central result.

H.1 Square-root case from scratch

For $r = 2$ and $z_n = e^{i\theta_n}$,

$$\theta_{n+1} = \begin{cases} (\theta_n + \pi)/2, & \theta_n \leq 0, \\ (\theta_n - \pi)/2, & \theta_n > 0. \end{cases}$$

Taking absolute values gives

$$\alpha_{n+1} = \frac{\pi - \alpha_n}{2}.$$

The fixed point is $\alpha_* = \pi/3$, and

$$\alpha_n - \frac{\pi}{3} = \left(-\frac{1}{2}\right)^n \left(\alpha_0 - \frac{\pi}{3}\right).$$

Hence the sign alternates and the limiting points are $e^{\pm i\pi/3}$.

H.2 Why the naive lifted recurrence fails

The incorrect lifted recurrence would be

$$\tilde{\theta}_{n+1} = \frac{\tilde{\theta}_n + \pi}{2}.$$

Its fixed point is π , giving a formal limit of -1 . But π is outside the chosen principal interval except as a boundary value, and the actual complex principal square root performs a principal-value reduction before division. Therefore the lifted recurrence is not the principal map.

H.3 A simple numerical sanity table

For $z_0 = 1$ the first values of the absolute angle are

$$0, \frac{\pi}{2}, \frac{\pi}{4}, \frac{3\pi}{8}, \frac{5\pi}{16}, \frac{11\pi}{32}, \dots$$

which converge to $\pi/3$.

Appendix I

Proof Audit for the Pulickal Framework

I.1 Dependency chain

$$F_S \implies \sigma_S \implies \sim_S \implies X / \sim_S \implies D(S) \implies \text{state separation} \implies \text{capacity impossibility}.$$

The approximate branch replaces the finite quotient by a metric signature space and uses packing/covering numbers.

I.2 Scope checklist

Before applying the framework to a new problem, verify:

1. What is an information state?
2. What future tasks are admissible?
3. Is the response deterministic?
4. What persistent information is included?
5. What external channels are allowed?
6. What architecture class is under study?
7. What resource budget is fixed?
8. What error metric is used in the approximate case?

If these questions are not answered, the resulting lower bound is likely to be a theorem about a narrower model than intended.

Appendix J

Exact Finite Computation Protocol for $M_k(X, H)$

The computation is included as mathematics, not merely as software advice.

J.1 Layer recursion

Given $A = A_k(H)$, define

$$E_0 = \{0\}, \quad E_h = E_{h-1} + A \cup E_{h-1} - A,$$

where

$$E \pm A = \{e \pm a : e \in E, a \in A\}.$$

Then E_h is exactly the set of sums of h signed atoms.

J.2 Exactness

There is no floating-point arithmetic. Every state is an integer. A bitset representation indexed over $[-hH, hH]$ is exact because no h -term sum leaves that interval.

J.3 Certification

A finite computation should be independently checked by at least one of the following:

1. a sparse-set implementation on small (X, H) ;
2. direct verification of every stored minimal representation;
3. a second independent implementation;
4. a proof that the stopping layer covers every target in the finite range.

Appendix K

Additional Counterexamples Worth Remembering

K.1 Amplitude normalization

$\|\psi\|_2 = 1$ does not imply $|\int_{\Omega} \psi \, d\mu|^2 = 1$.

K.2 Four-valued functional completeness

The subset $\{0, i\}$ is closed under the proposed NAND, preventing generation of 1 and -1 from those inputs.

K.3 Transform novelty

Replacing a transform parameter by a complex value is still ordinary complex analysis unless the transform space or operator class has been changed.

K.4 Physical universality

A mathematically elegant recurrence does not prove a universal model of lightning, finance, medicine, or noise. Such claims require independently specified physical systems, measurements, error models, and experiments.

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Final Notebook Note

A notebook becomes valuable when its pages make it easier to tell the difference between a theorem, a conjecture, a useful model, an attractive analogy, and an error. The permanent record is not only the successes. The corrections are part of the mathematics because they show exactly where an argument changed category.

For this notebook, the standard going forward is therefore deliberately simple: every new page should say what its objects are, what its assumptions are, what its strongest proved statement is, what remains conjectural, and what has been ruled out.

Define first. Branch carefully. Prove what is provable. Label the rest.

RESEARCH NOTES



For calculations, corrections, observations, and future extensions



This inaugural volume establishes the foundation of the Pulickal Framework. It provides a mathematically exact reconstruction of recursive complex roots, formally separating deterministic principal orbits from multivalued inverse geometries.

Moving beyond pure dynamics, the text introduces a rigorous task-semantic framework for information states—proving that a specification’s complexity is driven not by its textual length, but by the precise set of future behaviors that must remain distinguishable. Finally, it formulates the finite-height pure-prime-power representation problem, bridging algebraic geometry with sparse additive combinatorics.

A deliberate rejection of infinite regress in favor of exact boundaries between semantic necessity and physical cost.

INDEPENDENT RESEARCH MONOGRAPH

Self-Contained & Mathematically Exact